

The structures and meanings of *might*-counterfactuals: a view from Japanese*

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Abstract

Might-counterfactuals (e.g., *if John had been at the party, Mary might have been happier*) have been observed, primarily by philosophers, to have a range of interpretations that are apparently absent in the corresponding *would*-counterfactuals. This paper probes into the structures and meanings of *might*-counterfactuals from the perspective of Japanese, a language that can encode relative scope relations between tenses and modals more transparently than English does. The investigation begins with a survey of the discussion in the philosophical literature. There are three major logical forms proposed for *might*-counterfactuals, which I call One-Modal, Wide-Scope, and Narrow-Scope, distinguished by the way they define the relation between the possibility modal and the conditional connective. Using them as prototypes for my account, I propose a way to construct linguistic counterparts of these different analyses of *might*-counterfactuals and formulate predictions concerning how they are instantiated by different sentential forms of *might*-counterfactuals in Japanese. I propose a set of empirical diagnostics to test these predictions and apply them to Japanese *might*-counterfactuals. I finally discuss potential implications of my investigation for English *might*-counterfactuals and for the theory of conditionals in natural language.

Keywords: Counterfactuals, *Might*-counterfactuals, Metaphysical modality, Epistemic modality, Japanese

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1 Introduction

Counterfactuals (henceforth CFs) are for instance exemplified by sentences like (1a) and (1b), so-called *would*-CFs.¹

- (1) a. If John were at the party, Mary *would* be happier.
- b. If John had been at the party, Mary *would* have been happier.

Sentences such as (2a) and (2b), on the other hand, exemplify so-called *might*-CFs. In English, *might*-CFs are formally identical to *would*-CFs except for the modal auxiliary in the consequent.

- (2) a. If John were at the party, Mary *might* be happier.
- b. If John had been at the party, Mary *might* have been happier.

My central concern is this latter variant of CFs. *Might*-CFs have been particularly controversial in the philosophical literature. Despite their superficial resemblance to *would*-CFs, *might*-CFs have been observed to have a range of interpretations that are apparently absent in the corresponding *would*-CFs. A variety of logical forms have been proposed, each motivated on different semantic or philosophical grounds.

While *might*-CFs have been actively discussed in philosophy, there has been no work, to the best of my knowledge, that probes into their structures and meanings from a linguistic perspective.² Despite the scarcity of work so far, relevant issues deserve thorough empirical and theoretical investigation. If *might*-CFs do exhibit a wide range of interpretations, how do these interpretations arise from the corresponding linguistic structures? Are the intuitions reported by philosophers motivated (or unmotivated) linguistically? How do languages other than English express *might*-CFs and what cross-linguistic variation and uniformity will be observed? Answering these questions will not only impact the semantic theory of modality and conditionals, but also contribute to the long-standing philosophical debate.

In this paper, I tackle these issues by using Japanese as a case study. The crucial advantage of Japanese is its transparency concerning relative scope relations between tenses and modals. This is advantageous for the purpose of formulating and testing a compositional account, especially as *might*-CFs are predicted to involve multi-layered interactions between tenses and modals. Such transparency is lacking in English, as evidenced by the famous ambiguity of *might have* (see Condoravdi 2002; Groenendijk and Stokhof 1975; Rullmann and Matthewson 2018). Japanese, in this sense, offers us a vantage point that enables us to see a clearer empirical landscape.

This paper is structured as follows. Section 2 discusses how *might*-CFs have been analyzed by philosophers. Section 3 discusses how Japanese differs from English as to

¹I refer to CFs as those conditionals whose antecedents are known or believed to be false. In this paper I will not deal with, e.g., so-called Anderson conditionals (Anderson 1951), which look like CFs but are used to argue for the truth of the antecedent (e.g., *if Jones had taken arsenic, he would show just these symptoms which he is now showing*). See also Mizuno 2024 for discussion of Japanese Anderson conditionals.

²This is not to say that *might*-CFs have never been discussed in linguistics: Asher and McCready (2007) discussed relevant constructions in connection to modal subordination; Alonso-Ovalle (2009) dealt with English *might*-CFs that involve disjunctive antecedents; see also Abusch 2012; Greenberg 2021; Komoto 2010. My point is that there has been no previous linguistic work that tackled the compositional semantics of *might*-CFs head-on as an independent problem, drawing attention to the intricacies of the interaction between modal and temporal ingredients.

the way it encodes scope interactions between tenses and modals. Section 4 introduces formal and theoretical preliminaries for my compositional analysis. Section 5 shows predictions concerning what structures and interpretations are assigned to each variant of Japanese *might*-CFs. The predictions are tested against empirical data in Sections 6-8. Section 9 discusses implications of the present work. Section 10 concludes.

A remark on data collection: Japanese data from Sections 3 and 6-8 are the main empirical contribution of this paper. Each piece of data from these sections (and other sections occasionally) is accompanied by the distribution of judgments collected from four native speakers of Japanese, including myself (“ok” = acceptable, “?” = slightly off, “??” = significantly degraded, “#” = extremely degraded).³ For the three speakers other than myself, judgments were collected through informal elicitation. I consulted each speaker separately in person. For each sentence, I first showed the context and asked how acceptable the sentence was in that context. Comments from my consultants are provided where they become relevant.

2 What might *might*-CFs mean?

In philosophy, the semantics of *might*-CFs has been discussed as part of the so-called “Duality vs. Conditional Excluded Middle” debate.⁴ My aim here is not to adjudicate between the two sides, but to survey the variety of logical forms and interpretations highlighted in the course of the debate. The discussions by philosophers involve many useful observations and arguments that serve as our starting point.

There are three major logical forms proposed for *might*-CFs.⁵ Lewis (1973) proposed an analysis that amalgamates *might* and the conditional connective into a special two-place connector “ $\Diamond\rightarrow$ ”, which I call *One-Modal* in this paper. Couched in the possible worlds framework, $\phi\Diamond\rightarrow\psi$ is analyzed as conveying that at some ϕ -world most similar to the actual world, ψ is true. Lewis argues that, with this analysis, (3) is “plainly false” if there was no penny in the pockets in the actual world, because the most similar worlds where I looked in my pockets are all no-penny-in-the-pockets worlds.

(3) If I had looked in my pockets, I might have found a penny.

Stalnaker (1981) and DeRose (1994, 1999), however, claim that (3) is not plainly false.⁶ DeRose (1994) writes the following (I replace the conditional connective in the text with my own “ $\triangleright$ ”, which is intended as a theory-neutral two-place connective representing *would*-CFs):

“Suppose, that is, that in the example he still didn’t know whether there had been a penny in his pocket, and was therefore ignorant of the truth-values of $(\phi \triangleright \psi)$ and $(\phi \triangleright \neg\psi)$. In

³The speakers are all linguists in their twenties to thirties. Two of them are from the eastern region of Japan, one from the central region and one from the western region.

⁴Duality, famously endorsed by Lewis (1973), analyzes *might*-CFs as the negation of the corresponding *would-not*-CFs (i.e., *if ϕ , might ψ* $\Leftrightarrow \neg$ (*if ϕ , would $\neg\psi$*)). Conditional Excluded Middle, endorsed by Stalnaker (1968), dictates on the other hand that the negation of *would-not*-CFs is the corresponding *would*-CFs (i.e., “*if ϕ , would ψ* $\vee$ *if ϕ , would $\neg\psi$* ” is a logical truth). See, e.g., Mandelkern 2018 for recent discussion of the debate.

⁵See, e.g., Bennett 2003 for more detailed historical background.

⁶See also Schultheis 2024 for a recent defense of Wide-Scope.

this situation, . . . [our (3)] certainly could be a very natural and *appropriate* thing for him to say.” (DeRose 1994, p.414)

Stalnaker and DeRose claim that there are contexts in which *might* expresses speaker uncertainty regarding the truth value of a *would*-CF, and that such contexts call for a logical form in which *might* scopes over a *would*-CF. I call this analysis *Wide-Scope* and notate the relevant logical form as “ $\Diamond(\phi \triangleright \psi)$ ”, which conveys that there is some world accessible from the actual world at which $\phi \triangleright \psi$ is true. To support this view, DeRose provides the following scenario:

“A friend needed a penny some time ago and asked if you had one. You said you had none. And indeed you didn’t, but you didn’t know that you were pennyless; you denied having a penny because you were busy, your hands were full, and you didn’t want to bother searching your pockets for change.” (DeRose 1994, p.414)

Now suppose that the friend later came to suspect you and asked, “Would you have found a penny if you had looked in your pockets?”. Because you have bought several things in the meantime and thrust change into your pockets, you can no longer tell whether you actually had a penny. As DeRose observes, (3) can be felicitously uttered as a reply, despite the fact that you would have found none.

Lewis (1979, 1986) and Bennett (2003) suggested the possibility of *might* scoping inside the consequent of a *would*-CF. I call this *Narrow-Scope* and notate the relevant logical form as “ $\phi \triangleright \Diamond\psi$ ”, which conveys that at every ϕ -world most similar to the actual world, ψ is possible. Both Lewis and Bennett considered this logical form so close to One-Modal that it covers a largely or completely overlapping range of meanings.⁷ Bennett suggested that Narrow-Scope is equivalent to One-Modal and thus the latter can be dispensed with. Lewis claimed that they are not equivalent and thus *might*-CFs

⁷Their concrete justifications differ in a way that is irrelevant to the main concern of this paper. For more details, see each author’s work.

are ambiguous between the two.⁸ I will ultimately agree with Lewis that Narrow-Scope is not equivalent to One-Modal, though I suspect that One-Modal and Narrow-Scope do not diverge in the way Lewis predicted (see footnote 8). The relation between One-Modal and Narrow-Scope will be discussed extensively in Section 7.

We have thus seen three different analyses of *might*-CFs. The disagreement among philosophers reflects how diverse and elusive the semantics of this CF variant is.

Two additional remarks are in order. Firstly, to the best of my knowledge, no author has argued for the same sort of ambiguity for *would*-CFs. While I leave open whether one may find similar ambiguity for *would*-CFs, the fact that the relevant ambiguity has been noted exclusively for *might*-CFs suggests that there is something linguistically significant with *might*-CFs or the modal “*might*”, which is plausibly absent in *would*-CFs or the modal “*would*”. Secondly, the semantic variation of *might*-CFs that has been noted by philosophers does not exhaust all the analytical possibilities, especially regarding the flavor of *might*. This paper tries to cover the possibilities that have not been discussed by philosophers.

3 Starting with consequents

The last section introduced three logical forms for *might*-CFs: One-Modal “ $\phi \Diamond \rightarrow \psi$ ”, Wide-Scope “ $\Diamond(\phi \triangleright \psi)$ ” and Narrow-Scope “ $\phi \triangleright \Diamond \psi$ ”. They are prototypes for my investigation in the sense that they involve minimal assumptions about relations between conditional connectives and possibility modals. A linguistic analysis of *might*-CFs, however, needs to aim at a more fine-grained level of compositionality, so that interactions between modal and temporal ingredients can be captured correctly. To take on this task, I first consider the linguistic structures of the consequents, with an eye towards extending them into linguistic structures of *might*-CFs. My specific focus

⁸Lewis’s claim is based on his judgement of the following sentences. As he argues in Lewis 1973, a sequence like (i) gives rise to a feel of contradiction, which he attributes to the duality of *would*-CFs and *might*-CFs. Meanwhile, he argues in Lewis 1979, 1986 that there is an escape route: when (ib) is interpreted as in (iib), which he regards as a linguistic counterpart of Narrow-Scope, the sequence does not sound contradictory to him. According to Lewis, this is because Narrow-Scope conveys that at all the most similar ϕ -worlds, there is a chance of ψ : it is possible that ψ ends up false in every world, and when this happens it does not contradict the *would-not*-CF.

- (i) a. If it had been that ϕ , it would have been that $\neg\psi$.
 b. ... but if it had been that ϕ , it might have been that ψ .
- (ii) a. If it had been that ϕ , it would have been that $\neg\psi$.
 b. ... but if it had been that ϕ , there would have been some chance that ψ .

However, I hesitate to accept Lewis’s judgment of (i) and (ii) as it is. I asked two English speakers about (iia) and (iib). Contrary to Lewis’s intuition, they both found (iia) and (iib) equally contradictory, finding no escape route along the lines Lewis suggested.

- (iii) a. #If we had invited Chomsky to the colloquium, he would not have come; but if we had invited Chomsky to the colloquium, he might have come.
 b. #If we had invited Chomsky to the colloquium, he would not have come; but if we had invited Chomsky to the colloquium, there would have been some chance that he would (have) come.

I suspect that Lewis’s data do not distinguish between One-Modal and Narrow-Scope, at least linguistically. I claim that the two analyses can be disambiguated if we account for temporal factors, and for this to be achieved, we need a framework that incorporates the temporal dimension in addition to the modal dimension. I will elaborate on this in Section 7.

here is on scope interactions within English *might have* and its Japanese equivalents. I begin with a discussion of English and then turn to Japanese data.

3.1 English

Many previous works have noted that English *might have* exhibits systematic ambiguity with respect to its temporal interpretation (see [Condoravdi 2002](#); [Groenendijk and Stokhof 1975](#); [Rullmann and Matthewson 2018](#)). For instance, while the simple *might*-sentence (4a) conveys unambiguously that it *is* possible that John *will* win (i.e., looks towards the future from the present perspective), (4b) has at least two readings.⁹ On one reading, it conveys that it *is* possible that John *won* (i.e., looks towards the past from the present perspective). On another reading, it conveys that it *was* possible that John *would* win (i.e., looks towards the future from the past perspective).¹⁰

- (4) a. John might win.
b. John might have won.

The ambiguity of (4b) is generally analyzed as arising from different scope relations between *might* and *have* at the covert level (e.g., [Condoravdi 2002](#)). The first reading is analyzed as due to (4b) having the structure in (5a): the Perfect “*have*” appears between *might* and the sentence radical (i.e., a saturated verb phrase without any aspectually or temporally significant morphology) and shifts the reference time of the latter into the past; because *might* here outscopes *have*, its evaluation time (or, its perspective) is anchored to the utterance time. The second reading, on the other hand, is analyzed as arising from the structure in (5b): unlike in (5a), *might* here is outscoped by *have*, so that its evaluation time is shifted into the past; the reference time of the sentence radical is located in the future of the past time at which *might* is evaluated, due to the eventive aspect of the predicate.

- (5) a. [might [have [John win]]]
b. [have [might [John win]]]

If English *might have* is ambiguous and may have different underlying structures, one may reasonably predict that these structures are in some ways inherited by the structures of *might*-CFs, thus accounting for the interpretive variability of this CF variant. In fact, given a few standard linguistic assumptions, the structures in (5) can be extended into linguistic structures of *might*-CFs, in particular those that structurally correspond to One-Modal, Wide-Scope and Narrow-Scope.

Deferring a more detailed implementation until Section 4, here is a brief sketch. A linguistic analysis of CFs generally assumes that (i) *if*-clauses are restrictors of modals ([Kratzer 1986, 1991a, 2012](#)), and (ii) Past or Perfect in CFs enables access to worlds at which the antecedent is true (e.g., [Iatridou 2000](#); [Ippolito 2003, 2006, 2013](#)).

⁹As [Rullmann and Matthewson \(2018\)](#) noted, there is a third reading of (4b) which conveys that it *was* possible that John *won* (i.e., looks towards the past from the past perspective). This reading corresponds to the meaning of Japanese (10d) below.

¹⁰The discussion here leaves open the specific modal flavor of *might* in each sentence, though it is often noted that there is a correlation between temporal interpretations and salient flavors of modals. This will be discussed in Section 4.3.

Take the structure in (5b) for instance. Notice that, given the assumptions (i) and (ii), restricting *might* in this structure with an *if*-clause results directly in a possible linguistic structure of *might*-CFs. Informally speaking, *have* shifts the evaluation time of “ $\text{might}_{[\text{if } \phi]}$ ” into the past. This, together with the assumption that live possibilities are eliminated through the passage of time (Thomason 1984), triggers an expansion of the domain, enabling access to worlds at which the antecedent “ ϕ ” is true.¹¹ Notice that the structure created in this way, as shown in (6), can be considered a counterpart of One-Modal “ $\phi \Diamond \rightarrow \psi$ ”, because it involves just one modal in it.

$$(6) \quad [\text{have} [\text{might}_{[\text{if } \phi]} [\text{John win}]]] \quad (\text{One-Modal})$$

A counterpart of Wide-Scope “ $\Diamond(\phi \triangleright \psi)$ ” can be obtained from (5a), crucially by building on the additional assumption that a covert necessity modal may be inserted to get restricted by an *if*-clause (Kratzer 1986, 1991a, 2012). Specifically, by inserting a covert necessity modal “ $\Box$ ” below *have* and restricting it with an *if*-clause, (5a) can be extended into (7), in which *might* scopes over an entire *would*-CF.

$$(7) \quad [\text{might} [\text{have} [\Box_{[\text{if } \phi]} [\text{John win}]]]] \quad (\text{Wide-Scope})$$

A counterpart of Narrow-Scope “ $\phi \triangleright \Diamond \psi$ ” can be obtained from (5b) by building on the same assumption about covert necessity modals. Specifically, by inserting a covert necessity modal below *have* and restricting it by an *if*-clause, (5b) can be extended into (8), in which *might*, unlike in (7), scopes within the consequent of a *would*-CF.

$$(8) \quad [\text{have} [\Box_{[\text{if } \phi]} [\text{might} [\text{John win}]]]] \quad (\text{Narrow-Scope})$$

Thus, at least from a theoretical standpoint, the putative underlying structures of *might have* can be extended into possible linguistic structures of *might*-CFs, especially those that structurally correspond to the three logical forms from Section 2.

A natural question that would then arise is how much empirical validity structures like these have as an analysis of *might*-CFs in natural language. However, this would be difficult to test, or end up insufficiently addressed, as long as one focuses only on English. This is because the relevant structural differences are assumed to exist only at the *covert* level in this language: that is, *might*-CFs like (9) could map to any structure in (6)-(8) above, thus are ambiguous between a fairly wide range of meanings.

$$(9) \quad \text{If John had been in top shape, he might have won.}$$

Diagnostics that rely on judgments of felicity in context are thus hard to construct for English *might*-CFs: even if one tailored contexts that could rule out one interpretation, there would still be another interpretation that could make the sentence true in exactly the same contexts, thus rendering them ineffective as diagnostics. If English showed a more transparent form-meaning correspondence, linguistic investigation of *might*-CFs would be much easier to pursue.

¹¹This line of analysis of CFs is the so-called “Past-as-Past” analysis (see Arregui 2007, 2009; Ippolito 2003, 2006, 2013; Khoo 2015; Romero 2014; among others), in contrast to the “Past-as-Modal” analysis (see Iatridou 2000; Mackay 2015, 2019; Schulz 2014; among others). The choice of Past-as-Past here is mainly for presentational purposes. See Section 4.5 for further discussion.

3.2 Japanese

However, if there is a language that can do what English cannot, it will be an effective alternative probe for investigating *might*-CFs. I argue that Japanese is a language that is well-suited for this purpose, as it can express relative scope relations between modal and temporal elements more transparently than English does.

To illustrate, Japanese has a class of modals that must be overtly “sandwiched” by two tenses, exemplified by the possibility modal “*kamosirena*–”, which is glossed as ‘might’ in this paper. Because Japanese has two tenses (i.e., Non-Past and Past), there are four possibilities as to how the two tense slots are filled, as shown in (10).^{12,13}

- (10) a. [*John-ga kat*]-**u** *kamosirena-i*.
John-NOM win-NPST might-NPST
‘John might win.’ (≈ ‘It is possible that John will win.’)
- b. [*John-ga kat*]-**ta** *kamosirena-i*.
John-NOM win-PAST might-NPST
‘John might have won.’ (≈ ‘It is possible that John won.’)
- c. [*John-ga kat*]-**u** *kamosirena-katta*.
John-NOM win-NPST might-PAST
‘John might have won.’ (≈ ‘It was possible that John would win.’)
- d. [*John-ga kat*]-**ta** *kamosirena-katta*.
John-NOM win-PAST might-PAST
‘John might have won.’ (≈ ‘It was possible that John had won.’)

Notice that the outer tense determines the temporal perspective, whereas the inner tense determines the temporal orientation. In particular, (10b) corresponds to the first reading of English (4b) (i.e., looking towards the past from the present perspective), whereas (10c) corresponds to the second reading (i.e., looking towards the future from the past perspective).¹⁴ The difference between (10b) and (10c) becomes clearer when placed in appropriate contexts. In (11), the speaker makes a guess about a past event based on what she currently knows. In this context, only (11a) is felicitous.

¹² TOP = topic marker; NOM = nominative; GEN = genitive; DAT = dative; ACC = accusative; LOC = locative; CL = classifier; NEG = negation; ADV = adverb; GER = gerundive; COP = copula; ASP = aspectual marker; NPST = Non-Past; PAST = Past; SUBJ = subjunctive; COND = conditional connective; FIN = finite marker; Q = Q-particle; SFP = sentence final particle.

¹³ Japanese tense shows contextual allomorphy depending on whether the category of the predicate is a verb or an adjective. Non-Past is realized as *-(r)u* when it attaches to verbs, whereas it is realized as *-i* when it attaches to adjectives. Non-Past on *kamosirena*- surfaces as *-i* because this modal is formally categorized as an adjective. Past is realized as *-ta* and *-katta* when it attaches to verbs and adjectives respectively.

¹⁴ A reviewer of *L&P* finds sentences like (10c) somewhat stilted. As far as my consultants are concerned, no one found (10c) less natural than (10a) or (10b) (see Niwa 1992 for some discussion of the related issues). I also note that naturalistic occurrences of these constructions abound in the BCCWJ corpus (Maekawa et al. 2014), such as (i).

(i) *Zyuugeki-wa itumademo tudui-ta. Itunandoki, nagaredama-ni atar-u kamosirena-katta.*
shooting-TOP forever go.on-PAST at.any.moment stray.bullet-DAT be.hit-NPST might-PAST
‘Shooting went on forever. It was possible at any moment that I would get hit by a stray bullet.’
(BCCWJ: LBcn.00005)

- (11) (John is a boxer and there is a match in which he opposes the world champion. Caught in a traffic jam, the speaker fails to make it to the match, but when she finally arrives at the parking lot, she hears thunderous cheer coming from the arena, which she thinks suggests a surprising upset by John against the champion. She says to her company:)

[*Ima kikoete kuru ariina-no kansee-kara suruto, ...*]

now hear come arena-GEN cheer-from judging

‘Judging from the cheer from the arena we hear now, ...’

- a. ... *John-ga kat-ta (no) kamosirena-i.*

John-NOM win-PAST FIN might-NPST

‘... John might have won.’ (ok: 4/4)

- b. ... *John-ga kat-u (no) kamosirena-katta.*

John-NOM win-NPST FIN might-PAST

‘... John might have won.’ (#: 4/4)

Compare this with (12). Here the sentence-initial adverbial shifts one’s perspective into a past moment at which the presidential election in 2016 was to take place in one month: the event denoted by the sentence radical (i.e., Clinton winning the election by a wide margin) is a future event from the perspective of that moment. Here, only (12b) can be felicitously used. These data indicate that Japanese does encode relative scope relations between tenses and modals in a transparent way.

- (12) [*Senkyo-kara ikkagetu mae-no ziten-de-wa, ...*]

election-from one.month before-GEN moment-LOC-TOP

‘As the election was to take place in one month, ...’

- a. ... *Clinton-ga taisa-de kat-ta kamosirena-i.*

Clinton-NOM wide.margin-by win-PAST might-NPST

‘... Clinton might have won by a wide margin.’ (?: 1/4, #: 3/4)

- b. ... *Clinton-ga taisa-de kat-u kamosirena-katta.*

Clinton-NOM wide.margin-by win-NPST might-PAST

‘... Clinton might have won by a wide margin.’ (ok: 4/4)

This paper thus aims to tackle the semantics of *might*-CFs from the perspective of Japanese. My main concern is how (10a)-(10c) extend to different kinds of *might*-CFs and what interpretations they express. I exclude (10d), which involves two Pasts, from the scope of investigation, due to the complexity of the contribution that two Pasts make to the meaning of *might*-CFs, which I still do not fully understand and have to leave open.¹⁵ Leaving this for future work, I focus on the rest three forms.

In the remainder of this section, I clarify some more basics about the relevant Japanese constructions.

3.3 Some syntactic assumptions

In Japanese, as is fairly evident from the discussion of the previous section, the surface order of tenses and modals determines the structural hierarchy among these elements. Specifically, what is on the right is always structurally higher: I assume that there is no covert movement that changes this structural relation. It is thus assumed that (10a)-(10c) all have the structure in (13), and composition proceeds in a way that preserves this relation (I assume that sentence radicals correspond to *vP* in generative syntax).

- (13) [Outer tense [*kamosirena*- [Inner tense [*vP* Sentence radical]]]]

For conditionals, while the antecedent must surface sentence-initially in Japanese, I assume that, at the covert level, it figures in the position that adjoins immediately to the modal that it restricts. Addition of a conditional antecedent or a covert modal does not affect the structural relation in (13): whatever new element is inserted, outer tense outscopes *kamosirena*-, *kamosirena*- outscopes inner tense, and so on.

3.4 The modal flavors of *kamosirena*:- evidence from scope interactions

In the Japanese descriptive literature, *kamosirena*- is often classified into the category of modality that pertains to “likelihood” or “judgment of likelihood” (“*gaizensei*” or “*gaizensei handan*”; see, e.g., Moriyama 1989; Moriyama, Nitta, and Kudo 2000; Nitta 1989). *Kamosirena*- cannot be used to express prioritizing (e.g., deontic or bouletic) modality, for which Japanese has a reserved inventory of grammatical constructions (M. Kaufmann 2017; M. Kaufmann and Tamura 2020).

While there is consensus that *kamosirena*- can express an epistemic possibility, there has been no explicit discussion, to the best of my knowledge, of whether it can express a non-epistemic, *metaphysical* possibility (i.e., an objective chance for something to hold).^{16,17} Here I show that *kamosirena*- does have this latter construal, thus

¹⁵I would at least like to note that there are fairly many natural occurrences of *might*-CFs with this doubly Past-tensed form, e.g., (i).

(i) *Sinpu de ar-u koto-o simes-u eri-wa minituke-tei-na-i. Mosi ar-eba,*
 priest COP be-NPST fact-ACC indicate-NPST collar-TOP wear-ASP-be.not-NPST ADV be-COND
nanika-no yaku-ni tat-ta kamosirena-katta.
 something-GEN use-DAT stand-PAST might-PAST
 ‘He is not wearing a collar that indicates that he is a priest. If he had had one, that might have been of some use.’ (BCCWJ: LBF9.00033)

¹⁶The modality concerning “likelihood” in the descriptive literature is generally subsumed under the large category of “epistemic modality” (“*ninshiki no modaritii*”, which, though, covers more than what epistemic modality in formal semantics generally does, encompassing interrogatives, sentence final particles, etc.), thus as such *kamosirena*- has been treated as a kind of epistemic modal. More recent works like Larm 2006 and M. Kaufmann and Tamura 2020 also discuss *kamosirena*- in the realm of epistemic modality, though they seem to be open about whether *kamosirena*- has to be epistemically construed.

¹⁷I follow Condoravdi (2002) in calling such non-epistemic construals “metaphysical” construals, though they may also be identified with “circumstantial” construals in the literature. See Abusch 2012 for discussion of related issues.

it can have both epistemic and metaphysical flavors. Establishing this here is important because the variability of the flavor of the possibility modal between epistemic and metaphysical construals will play a role in distinguishing different kinds of *might*-CFs. The availability of metaphysical construals can be shown easily, by looking into the scope interaction with nominal quantifiers.

The idea is simple: (i) von Stechow and Iatridou (2003) observed that epistemic modals must outscope nominal quantifiers in the same clause, the constraint they dubbed the “Epistemic Containment Principle”; (ii) if *kamosirena-* can be outscoped by a nominal quantifier in the same clause, that indicates that this modal has metaphysical construals, in addition to epistemic construals.

To show this with a universal quantifier, consider a lottery for students where the first prize is awarded to only one of them (suppose that there are multiple students). This setup excludes the possibility that everyone ends up winning the first prize, thus excluding “ $\Diamond > \forall$ ” readings and forcing “ $\forall > \Diamond$ ” readings. Crucially, (14) is felicitous in such a context, confirming that the latter readings are available, which in turn indicates that *kamosirena-* has a metaphysical construal.¹⁸

- (14) (There is a lottery for students that awards the first prize to one randomly chosen person. The drawing takes place tomorrow. Everyone has an equal chance.)
 [Dono gakusee-ni-mo ittoo-ga atar]-u kamosirena-i.
 which student-DAT-also first.prize-NOM get-NPST might-NPST
 Lit. ‘Every student might win the first prize.’
 Intended: ‘For every student x , there is a chance that x wins the first prize.’
 (ok: 3/4, ? : 1/4)

This is further supported by the interaction with additive quantifiers. Suppose that John is involved in the same lottery. “ $\Diamond > \text{too}$ ” readings are excluded, as it is not possible for him and someone else to win simultaneously; thus, “ $\text{too} > \Diamond$ ” readings are forced. (15) can be felicitously used, thus *kamosirena-* is metaphysically construed.

- (15) (The same context as in (14))
 [John-ni-mo ittoo-ga atar]-u kamosirena-i.
 John-DAT-also first.prize-NOM get-NPST might-NPST
 Lit. ‘John, too, might win the first prize.’
 Intended: ‘John, too, has a chance of winning the first prize.’ (ok: 4/4)

One might argue that “ $\forall > \Diamond$ ” and “ $\text{too} > \Diamond$ ” readings are available because the Epistemic Containment Principle is not operative in Japanese. This is not correct. As noted in the literature, epistemic readings are forced when the prejacent (i.e., the proposition in the scope of a modal) is assumed to be “settled”—that is, its truth value is decided either way at the evaluation time for the modal (Condoravdi 2002;

¹⁸Some speakers (like my consultant who gave “?”) may find “ $\forall > \Diamond$ ” readings a bit hard to get: those who have difficulty are invited to insert a facilitating phrase like “*kooseena kuzi dakara, ...*” (‘because this is a fair lottery’) before the sentence. Note that this sort of facilitating phrase does not work for (16) and (17) below, suggesting that they do lack “ $\forall > \Diamond$ ” readings.

S. Kaufmann, Condoravdi, and Harizanov 2006).¹⁹ This means that the flavor of *kamosirena-* in (11), whose prejacent concerns the past, is obligatorily epistemic, while that in (14) and (15), whose prejacent concerns the future, is not. Now suppose that the same lottery as above has taken place but the speaker does not know the result, forcing *kamosirena-* to be epistemic. Unlike (14) and (15), neither of (16) and (17) can be felicitously used, indicating that the quantifier here cannot outscope *kamosirena-*, which in turn indicates that the Epistemic Containment Principle is in effect.

- (16) (There is a lottery for students that awards the first prize to one randomly chosen person. The drawing has already taken place, but the speaker doesn't know the result, though she knows everyone had an equal chance.)

[*Dono gakusee-ni-mo ittoo-ga atat*]-**ta** *kamosirena-i*.

which student-DAT-also first.prize-NOM get-PAST might-NPST

Lit. 'Every student might have won the first prize.'

Intended: 'For every student x , there is a chance that x won the first prize.'

(#: 4/4)

- (17) (The same context as in (16))

[*John-ni-mo ittoo-ga atat*]-**ta** *kamosirena-i*.

John-DAT-also first.prize-NOM get-PAST might-NPST

Lit. 'John, too, might have won the first prize.'

Intended: 'John, too, has a chance of having won the first prize.' (#: 4/4)

Given that the Epistemic Containment Principle is in effect, the availability of " $\forall > \Diamond$ " readings and "too $> \Diamond$ " readings in (14) and (15) respectively is more plausibly due to *kamosirena-* having non-epistemic construals.

Note furthermore that " $\forall > \Diamond$ " readings and "too $> \Diamond$ " readings are respectively available in (18) and (19). This is explained by the fact that the prejacent events here are future events from the perspective of the past (i.e., the truth values are not settled at the past moment), allowing for metaphysical construals.

- (18) (There was a lottery for students that would award the first prize to one randomly chosen person. The drawing was about to take place. Everyone had an equal chance.)

[*Dono gakusee-ni-mo ittoo-ga atar*]-**u** *kamosirena-katta*.

which student-DAT-also first.prize-NOM get-NPST might-PAST

Lit. 'Every student might have won the first prize.'

Intended: 'For every student x , there was a chance that x would win the first prize.' (ok: 4/4)

- (19) (The same context as in (18))

[*John-ni-mo ittoo-ga atar*]-**u** *kamosirena-katta*.

John-DAT-also first.prize-NOM get-NPST might-PAST

¹⁹This is because in such cases the sentence with a metaphysically construed possibility modal becomes semantically equivalent to its prejacent. The use of metaphysically construed modals thus requires the "Diversity Condition" to be satisfied by the context (Condoravdi 2002).

Lit. ‘John, too, might have won the first prize.’

Intended: ‘John, too, had a chance of winning the first prize.’ (ok: 4/4)

To summarize, *kamosirena*- has both epistemic and metaphysical interpretations. The scope interaction between modals and nominal quantifiers will also be used to distinguish different kinds of *might*-CFs in Section 6.

4 Setting the stage

In this section, I introduce formal and theoretical preliminaries for my compositional analysis of *might*-CFs. The reader who is primarily interested in the data and analysis of *might*-CFs may skip to Section 5. I adopt the $T \times W$ -frame from Thomason 1984, which models the interaction between temporal and modal dimensions in a systematic way. Many of the auxiliary notions are adopted from the compositional analysis of English indicative conditionals in S. Kaufmann 2005b, though some important peculiarities of Japanese are noted along the way. The full-fledged list of formal definitions is shown in Appendix A.

4.1 $T \times W$ -frame

The core intuition behind $T \times W$ -frames (Thomason 1984) is an asymmetry between fixed past and open future. The most fundamental notion is that of *historical alternatives*, which are modeled by $\approx$, a tertiary relation between two worlds and a time: “ $w \approx_t w'$ ” means that w and w' are historical alternatives at t . Intuitively, historical alternatives are worlds that share the same history up to some point. They form an equivalence class by assigning to any atomic sentences (here, sentence radicals) the same values up to t , but may depart from each other thereafter. This is illustrated in Figure 1, the “filter-funnel” model of historical necessity (S. Kaufmann 2005a; Thomason 1984). The temporal precedence is defined by a linear order “ $<$ ”.

In $T \times W$ -frames, sentences are interpreted with respect to world-time pairs. It will be helpful to extend $<$ and $\approx$ in the following way.

- (20) a. $\langle w, t \rangle < \langle w', t' \rangle$ iff $w = w'$ and $t < t'$.
 b. $\langle w, t \rangle \approx \langle w', t' \rangle$ iff $w \approx_t w'$ and $t = t'$.

I use $i, j, \dots$ for world-time pairs, or (Montagovian) indices: (20a) and (20b) are thus expressed as “ $i < j$ ” and “ $i \approx j$ ” respectively, with $i = \langle w, t \rangle$ and $j = \langle w', t' \rangle$.

I use $\phi, \psi, \dots$ for sentences. I use $\llbracket \cdot \rrbracket$ as a function which assigns each linguistic expression its intension. The intension of a sentence is a *proposition*, that is, a set of indices at which it is true (or a function from indices to truth values).

4.2 Tenses

I assume that tenses denote existential quantifiers over indices (Prior 1967; see also S. Kaufmann 2005b; Ogiwara 1996). Past and Non-Past are analyzed as encoding different temporal restrictions. (21) shows their interpretations (I use $p, q, \dots$ as variables

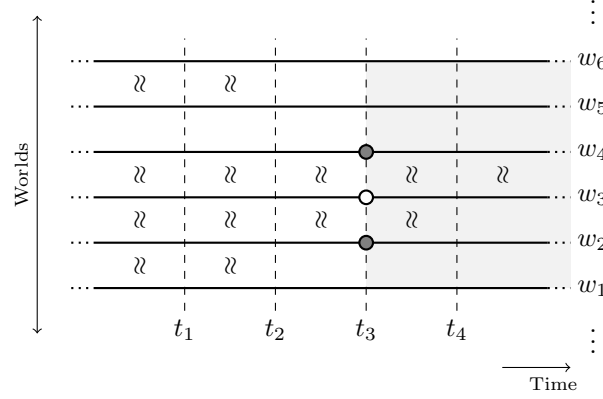


Fig. 1 The departing of worlds through time (slightly adapted from Mizuno and Kaufmann 2022). The rotated “ $\approx$ ” between world-lines indicates that the two worlds are historical alternatives up to the next vertical dashed line on the right. The historical alternatives of $\langle w_3, t_3 \rangle$ (the white dot) include itself and $\langle w_2, t_3 \rangle$ and $\langle w_4, t_3 \rangle$ (the dark gray dots). The gray area from t_3 onwards indicates that the future from then on is unforeseeable.

for propositions). (22) shows a sample derivation: the sentence is true at an utterance index i_u iff there is some index in the past of i_u at which John’s winning holds.

- (21) a. $\llbracket \text{PAST} \rrbracket = \lambda p \lambda i. \exists j < i. p(j)$
b. $\llbracket \text{NPST} \rrbracket = \lambda p \lambda i. \exists j \geq i. p(j)$
- (22) a. $[John\text{-}ga \quad kat]\text{-}ta.$
John-NOM win-PAST
b. $\llbracket \text{PAST} [John \text{ win}] \rrbracket$
 $= \llbracket \text{PAST} \rrbracket(\llbracket John \text{ win} \rrbracket)$
 $= \lambda i. \exists j < i. \llbracket John \text{ win} \rrbracket(j)$

4.3 Modals

I assume that modals are quantifiers over a domain of indices (see Kratzer 1981, 1991b, 2012; Yalcin 2007). Existential and universal quantification over a set of historical alternatives correspond to metaphysical possibility and necessity respectively.

In $T \times W$ -frames, the domain for epistemic modality is generally defined as the union of different sets of historical alternatives (Condoravdi 2002). This captures the idea of uncertainty about the past, under which one assumes that the event is settled but is uncertain which historical alternatives have turned out actual. The same idea is reproducible with an accessibility relation slightly different from $\approx$, which I adopt from S. Kaufmann 2005b: the epistemic accessibility relation “ $\sim$ ” is defined as a relation between indices that is transitive, serial and euclidean. $\sim$ defines a domain as the union of sets of historical alternatives if it *lacks foreknowledge* (S. Kaufmann 2005b): if an index can be reached through a relation that lacks foreknowledge, so can all of its historical alternatives. See Appendix A for formal details.

A domain of modal quantification at i is defined as a set of indices that are accessible from i through an accessibility relation R , which varies between $\approx$ and $\sim$. I use $\mathcal{D}^R(i)$ to refer to the domain of modal quantification at i determined by an accessibility relation R , that is, $\mathcal{D}^R(i) = \{j \mid iRj\}$.

Possibility ($\Diamond$) and necessity ($\Box$) modals are interpreted as follows.²⁰

- (23) a. $\llbracket \Diamond^R \rrbracket = \lambda p \lambda i. \exists j \in \mathcal{D}^R(i). p(j)$
b. $\llbracket \Box^R \rrbracket = \lambda p \lambda i. \forall j \in \mathcal{D}^R(i). p(j)$

The sentence (24a) is interpreted as in (24b): it is true at i_u iff there is an index j which is no earlier than i_u such that there is an index accessible from j such that John’s winning holds somewhere in its past. However, while the domain of the Non-Past semantically extends into the future of i_u , this future extension is generally not utilized for the Non-Past on modals, plausibly for pragmatic reasons.²¹ Formally, the temporal domain is restricted to the singleton “ $\{t_u\}$ ”, thus the Non-Past is in effect semantically inert. With this restriction in force, the interpretation of (24a), given in (24b), is equivalent to the simpler formula in (24c) (which is not the interpretation of a well-formed sentence). I will occasionally make use of this contextual equivalence, where it is warranted, to simplify the formulas.

- (24) a. $\llbracket [John-ga \quad kat]-ta \quad kamosirena \rrbracket -i.$
John-NOM win-PAST might-NPST
b. $\llbracket NPST [\Diamond^R [PAST [John \text{ win }]]] \rrbracket$
 $= \lambda i. \exists j \geq i. \exists k \in \mathcal{D}^R(j). \exists l < k. \llbracket John \text{ win } \rrbracket(l)$
c. $\llbracket \Diamond^R [PAST [John \text{ win }]] \rrbracket$
 $= \lambda i. \exists j \in \mathcal{D}^R(i). \exists k < j. \llbracket John \text{ win } \rrbracket(k)$

The sentence (25a) is interpreted as in (25b): it is true at i_u iff there is an index j in the past of i_u such that among those accessible from j , there is an index in whose future John’s winning holds.

- (25) a. $\llbracket [John-ga \quad kat]-u \quad kamosirena \rrbracket -katta.$
John-NOM win-NPST might-PAST
b. $\llbracket PAST [\Diamond^R [NPST [John \text{ win }]]] \rrbracket$
 $= \lambda i. \exists j < i. \exists k \in \mathcal{D}^R(j). \exists l \geq k. \llbracket John \text{ win } \rrbracket(l)$

To briefly note the interaction between temporal interpretations and modal flavors, R in (24) and (25) is to be instantiated by $\approx$ or $\sim$, but the choice is subject to several constraints. As noted in Section 3.4, only epistemic construals make sense when the prejacent is settled. Thus, in (24), $\sim$ is chosen.

For (25), while epistemic construals are often said to be unavailable when a modal is in the scope of Past (e.g., Condoravdi 2002; Groenendijk and Stokhof 1975), there are actually cases in which the modal in the scope of Past does seem to have epistemic construals, such as (26) (von Fintel and Gillies 2008; see also Chen et al. 2017; Rullmann and Matthewson 2018). In (26), the response to the *why*-question pertains

²⁰To avoid the clutter, I use the same symbols of accessibility relations in both the object language and the meta-language. I thank a reviewer of *L&P* for suggesting this.

²¹See Section 8 for relevant discussion.

to one's *past epistemic state*: even if there was no ice cream, the presence of some was epistemically conceivable at that moment. Such past epistemic readings are also possible in Japanese, but only when Past outscopes *kamosirena*-.

- (26) A: Why did you open the freezer?
 B: Because I might have found ice cream.
 B': *Aisu-o mituke-**ru** kamosirena-**katta** kara.*
 ice.cream-ACC find-NPST might-PAST because (ok: 4/4)
 B'': *Aisu-o mituke-**ta** kamosirena-**i** kara.*
 ice.cream-ACC find-PAST might-NPST because (? : 1/4, #: 3/4)

As noted in the literature, the range of environments in which past epistemic readings are available seems fairly limited, which has led some authors (e.g., [Hacquard 2010](#)) to cast doubt on the possibility of Past scoping over epistemic modals. In this paper, I follow [Rullmann and Matthewson \(2018\)](#) in assuming that Past *can* scope over epistemic modals, and the purported dispreference of past epistemic readings is mainly due to pragmatic factors (see [Rullmann and Matthewson 2018](#) for a battery of arguments against [Hacquard's](#) claim).²² The discussion here will become relevant in Section 8.

4.4 Conditionals

I assume that conditionals involve a modal that quantifies over a domain of indices that is restricted by the antecedent (see [Kratzer 1986, 1991a, 2012](#) among others). According to Kratzer's analysis ([Kratzer 1986, 1991a, 2012](#)), when the consequent of a conditional involves an overt modal, that overt modal is restricted by the antecedent; when the consequent involves no overt modal, a covert necessity modal " $\Box$ " is inserted to get restricted by the antecedent. However, whether insertion of a covert modal is *disallowed* in the presence of an overt modal is an open question (see, e.g., [Geurts 2004](#)), as is whether a covert modal is always *required* even in the presence of an overt modal (see, e.g., [Frank 1996](#)).²³ Here I take a neutral stance by assuming that an *if*-clause can restrict an overt modal, and even if the sentence already involves an overt modal, a covert modal can be inserted and get restricted by an *if*-clause. The issues here will be revisited in Section 9.2.

As noted in Section 3.3, I assume that a conditional antecedent syntactically adjoins to the modal it restricts. (27) illustrates the interpretations for such a complex.

- (27) a. $\llbracket \Diamond_{\text{IF-}\phi}^R \rrbracket = \lambda p \lambda i. \exists j \in \mathcal{D}^R(i) \cap \llbracket \phi \rrbracket. p(j)$
 b. $\llbracket \Box_{\text{IF-}\phi}^R \rrbracket = \lambda p \lambda i. \forall j \in \mathcal{D}^R(i) \cap \llbracket \phi \rrbracket. p(j)$

²²It seems still debatable whether the putative dispreference of past epistemic readings is a language-general phenomenon or just specific to English *might*. [Rullmann and Matthewson \(2018\)](#) suspect that the dispreferred status of past epistemic readings for English *might* may be due to the absence of overt marking for tense, suggesting that the same dispreference does not seem to arise for corresponding modal verbs in Dutch (which are freely inflected for tense) or semi-modals like *have to*. Indeed, none of my consultants had any difficulty with the past epistemic reading of *kamosirena*- in B's utterance in (26). An investigation of the general acceptability of past epistemic readings, however, is beyond the scope of this paper.

²³See also [Kratzer 2012](#), Ch.4 for a related discussion.

To briefly introduce some basics about Japanese conditionals, Japanese has a rich inventory of conditional connectives (see Arita 2007, 2009; Masuoka 1993 among others; see also Takubo 2020 for a recent overview). This paper focuses on *-ba*, which is one of the most commonly found conditional connectives in Japanese and can be used for both CF and non-CF conditionals.

An important feature of *-ba* is that it directly attaches to sentence radicals, precluding the possibility of tense appearing below it. For instance, in (28), the antecedent is formed by attaching *-ba* (here, one of its allomorphs “*-reba*”) directly to the sentence radical “*John-ga asita ku*” (‘John come tomorrow’).

- (28) [John-ga asita ku]-reba, [paatii-wa moriagar]-u.
 John-NOM tomorrow come-COND party-TOP be.fun-NPST
 ‘If John comes tomorrow, the party will be fun.’

Notice that the antecedent of (28) can make future reference despite the absence of any explicit operator that enables it in the same clause. This cannot be captured by simply applying (27), because then the domain would only include indices that are co-temporal to the utterance index. In order for the domain to include future indices, I assume following S. Kaufmann (2005b) that future reference of conditionals may be enabled by *forward-extension* of accessibility relations, which is attributed to the lexical meaning of the conditional connective.

- (29) $\llbracket \Box_{\text{IF}}^{R*} [John\ come\ tomorrow] \rrbracket [NPST [party\ be\ fun]]$
 $= \lambda i. \forall j \in \mathcal{D}^{R*}(i) \cap \llbracket John\ come\ tomorrow \rrbracket. \exists k \geq j. \llbracket party\ be\ fun \rrbracket(k)$

Forward-extension, indicated by the superscript “*” on R , allows one to access indices “diagonally” from the evaluation index i , covering both co-temporal and subsequent moments at all worlds accessible from i via R : together with the restriction by the antecedent, it shifts the temporal perspective to a future moment at which the truth value of the antecedent is already settled (see Figure 2). See Appendix C for further justifications for assuming forward-extension rather than covert tense.

The final assumption I introduce for the semantics of conditionals is *inheritance of antecedent restriction*. This I assume is a natural constraint to apply when there is an additional modal in the consequent: thus, in (30), only (30a) is allowed.

- (30) a. $\llbracket \Box_{\text{IF}-\phi}^R [\Diamond_{\text{IF}-\phi}^R [NPST\ \psi]] \rrbracket$
 $= \lambda i. \forall j \in \mathcal{D}^R(i) \cap \llbracket \phi \rrbracket. \exists k \in \mathcal{D}^R(j) \cap \llbracket \phi \rrbracket. \exists l \geq k. \llbracket \psi \rrbracket(l)$
 b. $\llbracket \Box_{\text{IF}-\phi}^R [\Diamond^R [NPST\ \psi]] \rrbracket$
 $= \lambda i. \forall j \in \mathcal{D}^R(i) \cap \llbracket \phi \rrbracket. \exists k \in \mathcal{D}^R(j). \exists l \geq k. \llbracket \psi \rrbracket(l)$

Without this constraint, the domain for a lower modal (here, $\Diamond^R$ in (30)) could include indices at which the antecedent does not hold (see A.S. Gillies 2010 for an array of

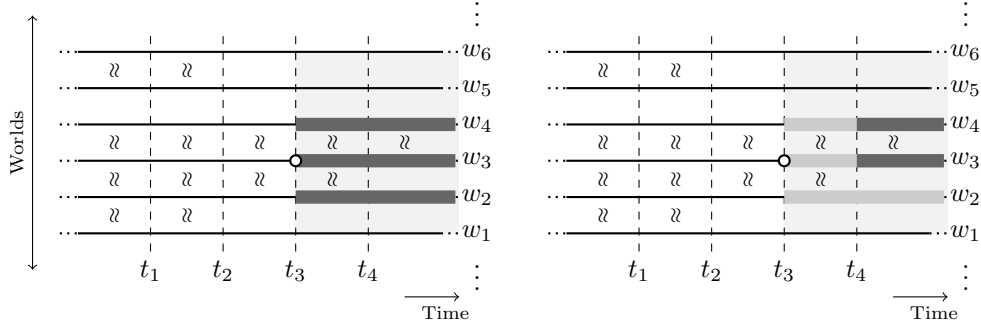


Fig. 2 The depiction of forward-extension. With forward-extension, the domain can include any index on the three thick dark gray horizontal lines that extend into the future from each of the historical alternatives of $\langle w_3, t_3 \rangle$ (Left). In evaluating a conditional, the domain is restricted by the antecedent proposition, which, here, results in the indices on the two thick dark gray lines extending from $\langle w_3, t_4 \rangle$ and $\langle w_4, t_4 \rangle$ (Right).

arguments for this sort of constraint).²⁴ Inheritance of restriction will be relevant to the interpretation of Narrow-Scope *might*-CFs.

4.5 Counterfactuals

Here I first introduce the basics of Japanese CFs and then illustrate a compositional analysis of Japanese CFs.

²⁴Here is an example that argues for assuming an inheritance constraint. In (i) and (ii), the stacked modals collapse to the innermost one, here $\diamond^{\sim}$, but while the restriction by ϕ is preserved in (i), it is lost in (ii). (i) and (ii) lead to different truth conditions: (i) is false at i if there is no index in $\mathcal{D}^{\sim}(i) \cap \llbracket \phi \rrbracket$ at which NPST ψ is true, but (ii) may be true in the same situation, as long as there is an index in $\mathcal{D}^{\sim}(i)$ that is a witness of NPST ψ . Obviously the interpretation in (i) better represents the semantics of conditionals.

- (i) $\llbracket \Box_{\text{IF-}\phi}^{\sim} [\diamond_{\text{IF-}\phi}^{\sim} [\text{NPST } \psi]] \rrbracket$
 $= \lambda i. \forall j \in \mathcal{D}^{\sim}(i) \cap \llbracket \phi \rrbracket. \exists k \in \mathcal{D}^{\sim}(j) \cap \llbracket \phi \rrbracket. \exists l \geq k. \llbracket \psi \rrbracket(l)$
 $\Rightarrow \lambda i. \exists j \in \mathcal{D}^{\sim}(i) \cap \llbracket \phi \rrbracket. \exists k \geq j. \llbracket \psi \rrbracket(k)$
- (ii) $\llbracket \Box_{\text{IF-}\phi}^{\sim} [\diamond^{\sim} [\text{NPST } \psi]] \rrbracket$
 $= \lambda i. \forall j \in \mathcal{D}^{\sim}(i) \cap \llbracket \phi \rrbracket. \exists k \in \mathcal{D}^{\sim}(j). \exists l \geq k. \llbracket \psi \rrbracket(l)$
 $\Rightarrow \lambda i. \exists j \in \mathcal{D}^{\sim}(i). \exists k \geq j. \llbracket \psi \rrbracket(k)$

A relevant constraint is called “shiftiness” in A.S. Gillies 2010. The analysis by Gillies is couched in what he calls the “operator” view on conditionals, but it can be largely spelled out with the apparatus of the restrictor analysis. See M. Kaufmann and Kaufmann 2015.

A reviewer for *L&P* suggested that the inheritance constraint may not generally hold for all conditionals, offering (iii):

- (iii) If he is here, he shouldn’t be.

As the reviewer pointed out, (iii) is a perfectly good thing to say but would not make sense if the modal in the consequent in (iii) inherited the antecedent restriction. I suspect that (iii) is interpreted analogously to “biscuit conditionals” (e.g., *there are biscuits on the sideboard if you want them*; Austin 1961). For instance, (iii) is odd when the consequent involves *then* (??*If he is here, then he shouldn’t be*; I thank Si Kai Lee for the judgment), which is a characteristic property of biscuit conditionals (see, e.g., Ebert, Ebert, and Hinterwimmer 2014; Iatridou 1991). Leaving further discussion for another occasion, I assume that the inheritance constraint is operative at least for the ordinary conditionals.

4.5.1 Basics about Japanese CFs

In many languages, counterfactuality is signaled by additional layers of Past or Perfect morphology, which have been dubbed *Fake Past* (Iatridou 2000).²⁵ As shown in (31), English has two varieties, Simple Past CFs (31a) and Past Perfect CFs (31b), differing in the number of layers of Fake Past involved (see also Ippolito 2003, 2006, 2013).

- (31) a. If John *came* tomorrow, the party *would* be fun.
 b. If John *had come* tomorrow, the party *would have* been fun.

Previous works on Fake Past are classified by and large into the “Past-as-Modal” analysis (Iatridou 2000; Mackay 2015, 2019; Ogihara 2014; Schulz 2014; Starr 2014; among others) and the “Past-as-Past” analysis (Arregui 2007, 2009; Ippolito 2003, 2006, 2013; Khoo 2015; among others). The Past-as-Modal analysis argues that Fake Past expresses “remoteness” from actuality, just as the ordinary use of Past expresses remoteness from the present moment. The Past-as-Past analysis argues that Fake Past expresses “temporal backshift”, re-running history from a time at which the antecedent was still contingent.

Like English, Japanese has Fake Past (Mizuno and Kaufmann 2019; Ogihara 2014), but Fake Past in Japanese has been noted to force the meaning equivalent to English Past Perfect CFs. The meaning equivalent to English Simple Past CFs is conveyed by the same form as indicative conditionals.²⁶

(32) illustrates a basic paradigm. (32b) is a CF with Fake Past: Past in the consequent is fake, as is clear from the mismatch with the future reference. (32a) is a CF without Fake Past. Its sentential form is identical to the indicative conditional in (28), indicating that it is ambiguous between the indicative and CF meanings. Counterfactuality, however, can be optionally signaled by the sentence final particle “*noni*”, which can also be used for CFs with Fake Past.²⁷

- (32) a. [John-ga asita ku]-reba, [paatii-wa moriagar]-u (noni).
 John-NOM tomorrow come-COND party-TOP be.fun-NPST SFP
 ‘If John *came* tomorrow, the party *would* be fun.’
 b. [John-ga asita ku]-reba, [paatii-wa moriagar]-ta (noni).
 John-NOM tomorrow come-COND party-TOP be.fun-PAST SFP
 ‘If John *had come* tomorrow, the party *would have* been fun.’

I show two pieces of evidence that suggest parallelisms between English (31a) and Japanese (32a) on the one hand, and English (31b) and Japanese (32b) on the other.²⁸ First, past reference of the antecedent is possible only in the latter class.

- (33) a. #[John-ga kinoo ku]-reba, [paatii-wa moriagar]-u (noni).
 John-NOM yesterday come-COND party-TOP be.fun-NPST SFP
 #If John came yesterday, the party would be fun.

²⁵ von Fintel and Iatridou (2023) have recently proposed the new term “X-marking” to cover a variety of morphological items that are used for CF expressions, including Fake Past. Here I stick to using Fake Past.

²⁶ See also Akatsuka 1985 for a similar remark.

²⁷ See Takubo 2020 for recent discussion of the semantic contribution of *noni* in conditionals.

²⁸ The discussion here is based on Mizuno and Kaufmann 2019.

- b. [John-ga kinoo ku]-reba, [paatii-wa moriagat]-**ta** (noni).
 John-NOM yesterday come-COND party-TOP be.fun-PAST SFP
 ‘If John had come yesterday, the party would have been fun.’

Second, Ippolito (2003, 2006, 2013) observes that when the existence presupposition of an event participant in the antecedent is not satisfied at the utterance time, Past Perfect CFs must be used. Similarly in Japanese, only CFs with Fake Past are felicitous in such contexts, as shown in (34).²⁹

- (34) (John got involved in an accident and killed yesterday. He was supposed to attend tomorrow’s reunion, where his classmates would have seen him for the first time in years. One of the classmates says:)
- a. #Zan’nen da. [Kare-ga asita ku]-reba, [min’na yorokob]-**u**
 pity COP he-NOM tomorrow come-COND everyone be.glad-NPST
 noni.
 SFP
 # ‘What a pity. If he *came* tomorrow, everyone *would* be glad.’
- b. Zan’nen da. [Kare-ga asita ku]-reba, [minna yorokon]-**da**
 pity COP he-NOM tomorrow come-COND everyone be.glad-PAST
 noni.
 SFP
 ‘What a pity. If John *had come* tomorrow, everyone *would have* been glad.’

Concerning the semantics of Japanese Fake Past, I assume following Mizuno and Kaufmann (2019) that it supports the analysis of Past-as-Past. Evidence comes from counter-mathematicals in (35), which cannot be expressed by CFs with Fake Past in Japanese (they cannot be expressed by English Past Perfect CFs either; see Dudman 1984). (35b) in fact misleadingly implies that there was a moment at which nine could have been prime, which is at odds with our assumption that mathematical properties are timeless. The oddness of (35b) falls out if Japanese Fake Past enforces temporal backshift, which presupposes the existence of such a branching moment.³⁰

- (35) a. [Kyu-u-ga sosuu de ar]-eba, [san-de warikir-e-na]-**i**.
 nine-NOM prime COP.GER be-COND three-by divide-able-NEG-NPST
 ‘If nine were prime, it would not be divisible by three.’
- b. ??[Kyu-u-ga sosuu de ar]-eba, [san-de warikir-e-na]-**katta**.
 nine-NOM prime COP.GER be-COND three-by divide-able-NEG-PAST
 ??‘If nine had been prime, it would not have been divisible by three.’

²⁹The contrast in (34) falls out from the Past-as-Past analysis which I will introduce shortly. Informally put, Fake Past in (34b) rewinds time to a moment at which it was still possible for John to come tomorrow, concomitantly shifting the evaluation time of the existential presupposition into the past; the presupposition then does not have to be satisfied at the utterance time, hence the felicity of (34b). On the other hand, the absence of Fake Past forces the evaluation time of the existential presupposition to be fixed at the utterance time, hence the infelicity of (34a). See Mizuno and Kaufmann 2019 for more discussion.

³⁰This example argues against the Past-as-Modal approach to Japanese Fake Past (Ogihara 2014): if Past encoded remoteness from the actual world, the oddness of (35b) would not arise. See Mizuno and Kaufmann 2019 for more detailed discussion.

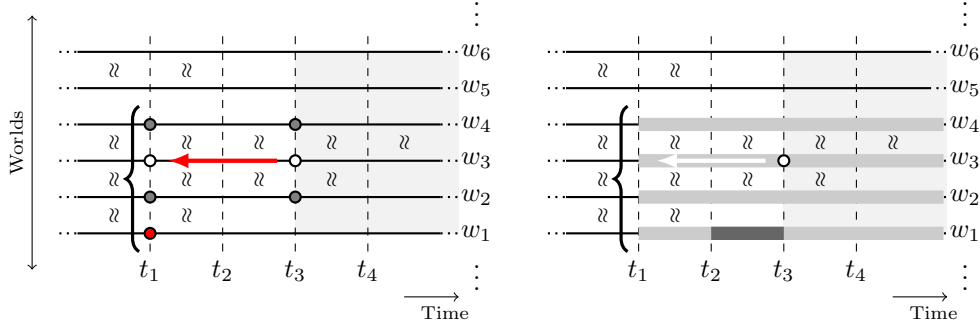


Fig. 3 The depiction of temporal backshift. Backshift of the time to t_1 enables access to w_1 , which is not a historical alternative of w_3 at t_3 (Left). Forward-extension then applies at t_1 , allowing one to access indices lying in the past or future of t_3 (Right).

To summarize: (i) Japanese has two types of CFs, namely, CFs with and without Fake Past: the latter share the same form with indicative conditionals, but the CF meaning can be forced by the particle “*noni*”. (ii) CFs with Fake Past semantically correspond to English Past Perfect CFs. (iii) Fake Past in Japanese supports the analysis of Past-as-Past, namely, temporal backshift. I henceforth call CFs with Fake Past *backshifting CFs*, and CFs without Fake Past *non-backshifting CFs*.

4.5.2 The semantics of Japanese CFs

I first show the analysis of backshifting CFs. The sentence (36a) is interpreted as in (36b).³¹ Past outscopes the conditional to enforce temporal backshift (see Figure 3). I assume that the modal restricted by a CF antecedent is primarily interpreted metaphysically (see Edgington 2007; Khoo 2015; Schulz 2007 for discussion of epistemic interpretations), and the conditional involves covert Non-Past under the modal, which allows forward-shifting for the consequent in cases like this (see Khoo 2015).

- (36) a. [John-ga asita ku]-reba, [paatii-wa moriagat]-ta.
 John-NOM tomorrow come-COND party-TOP be.fun-PAST
 b. $\llbracket \text{PAST} [\Box_{\text{IF}}^* [J \text{ come tomorrow}] \text{ NPST} [\text{party be fun}]] \rrbracket$
 $= \lambda i. \exists j < i. \forall k \in \mathcal{D}^*(j) \cap \llbracket J \text{ come tomorrow} \rrbracket. \exists l \geq k. \llbracket \text{party be fun} \rrbracket(l)$

For non-backshifting CFs, I assume that they involve what I call *vertical expansion*, which is indicated by the subscript “+” on R . Vertical expansion allows the domain to include co-temporal indices that are not historical alternatives of i (see Appendix A for formal details). The difference from temporal backshift is that it does not incorporate past indices (see Figure 4), which captures the fact that non-backshifting CFs are

³¹The evaluation of CFs is also subject to a similarity ordering of worlds or to causal dependence (Goodman 1947; Lewis 1973; Pearl 2000; Stalnaker 1968; among others). I assume that relevant restrictions are imposed on the quantificational domains of modals restricted by the antecedent (note, though, that these concerns are not central to the main focus of this paper).

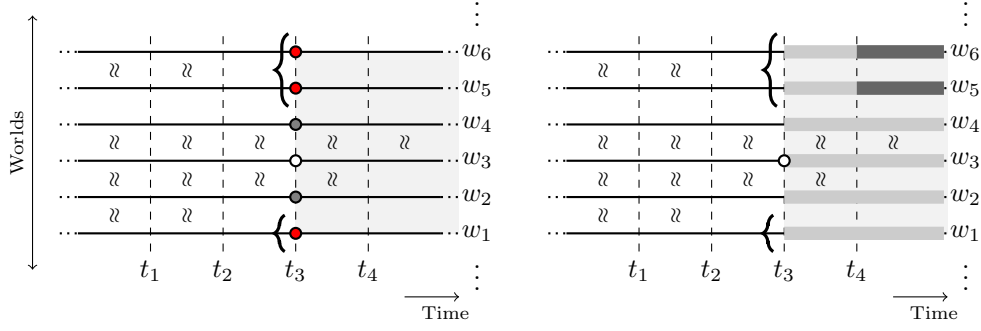


Fig. 4 The depiction of vertical expansion. It allows the domain to include indices that are co-temporal to, but not historical alternatives of, $\langle w_3, t_3 \rangle$ (Left). Forward-extension applies at t_3 , so the domain can include indices lying in the future of t_3 , but not those in the past of t_3 (Right).

incompatible with an antecedent about the past. (37) introduces the interpretation of the relevant conditionals. The sentence (38a) is interpreted as in (38b).

- (37) a. $\llbracket \Diamond_{\text{IF-}\phi}^{R_+} \rrbracket = \lambda p \lambda i. \exists j \in \mathcal{D}_+^R(i) \cap \llbracket \phi \rrbracket. p(j)$
b. $\llbracket \Box_{\text{IF-}\phi}^{R_+} \rrbracket = \lambda p \lambda i. \forall j \in \mathcal{D}_+^R(i) \cap \llbracket \phi \rrbracket. p(j)$
- (38) a. *[John-ga asita ku]-reba, [paatii-wa moriagar]-u (noni).*
John-NOM tomorrow come-COND party-TOP be.fun-NPST SFP
b. $\llbracket \Box_{\text{IF-}[John\ come\ tomorrow]}^{\approx_+^*} \rrbracket \llbracket \text{NPST } [party\ be\ fun] \rrbracket$
 $= \lambda i. \forall j \in \mathcal{D}_+^{\approx^*}(i) \cap \llbracket John\ come\ tomorrow \rrbracket. \exists k \geq j. \llbracket party\ be\ fun \rrbracket(k)$

This concludes the introduction of formal and theoretical preliminaries. In the next section, I will formulate predictions for the semantics of Japanese *might*-CFs.

5 Formulating predictions for Japanese *might*-CFs

Let me recapitulate the discussion from Section 3. English *might have* is ambiguous: one may look towards the past from the present, or towards the future from the past. This has been analyzed as due to different scope relations, (39a) for the former reading and (39b) for the latter reading.

- (39) John might have won.
a. $\llbracket \text{might } [\text{have } [\text{John win}]] \rrbracket$
b. $\llbracket \text{have } [\text{might } [\text{John win}]] \rrbracket$

One may hypothesize that these different structures of *might have* account for the semantic variation of *might*-CFs in English. With reasonable assumptions, (39a) and (39b) can extend to different structures of *might*-CFs, especially those corresponding to One-Modal “ $\phi \Diamond \rightarrow \psi$ ”, Wide-Scope “ $\Diamond(\phi \triangleright \psi)$ ” and Narrow-Scope “ $\phi \triangleright \Diamond \psi$ ”.

- (40) a. [have [might_[if ϕ] [John win]]] (One-Modal)
 b. [might [have [$\square$ _[if ϕ] [John win]]]] (Wide-Scope)
 c. [have [$\square$ _[if ϕ] [might [John win]]]] (Narrow-Scope)

However, the question of how valid these logical forms are as an analysis of *might*-CFs is hard to examine, since in English one sentential form could map to different underlying structures. An effective investigation calls for a language that can encode scope relations between modals and tenses more transparently. As I have been arguing, this is why Japanese offers a good testing ground. Recall that the two readings of *might have* are conveyed respectively by (41b) and (41c).

- (41) a. [John-ga kat]-**u** kamosirena-**i**.
 John-NOM win-NPST might-NPST
 ‘John might win.’ ($\approx$ ‘It is possible that John will win.’)
 b. [John-ga kat]-**ta** kamosirena-**i**.
 John-NOM win-PAST might-NPST
 ‘John might have won.’ ($\approx$ ‘It is possible that John won.’)
 c. [John-ga kat]-**u** kamosirena-**katta**.
 John-NOM win-NPST might-PAST
 ‘John might have won.’ ($\approx$ ‘It was possible that John would win.’)

Now, as all the formal and theoretical preliminaries have been introduced, I turn to formulating predictions concerning the form-meaning mapping in Japanese *might*-CFs.

(42) illustrates the paradigm of *might*-CFs that I will investigate (ϕ and ψ are sentence radicals). (43) shows concrete instantiations.

- (42) a. ϕ -COND, ψ -NPST kamosirena-NPST
 b. ϕ -COND, ψ -PAST kamosirena-NPST
 c. ϕ -COND, ψ -NPST kamosirena-PAST
- (43) a. [John-ga ku]-reba, [paatii-wa motto moriagar]-**u** kamosirena-**i**
 John-NOM come-COND party-TOP more be.fun-NPST might-NPST
 (*noni*).
 SFP
 ‘If John came, the party might be more fun.’
 b. [John-ga ku]-reba, [paatii-wa motto moriagar]-**ta** kamosirena-**i**
 John-NOM come-COND party-TOP more be.fun-PAST might-NPST
 (*noni*).
 SFP
 ‘If John had come, the party might have been more fun.’
 c. [John-ga ku]-reba, [paatii-wa motto moriagar]-**u** kamosirena-**katta**
 John-NOM come-COND party-TOP more be.fun-NPST might-PAST
 (*noni*).
 SFP
 ‘If John had come, the party might have been more fun.’

I first classify these as non-backshifting or backshifting. (42a) is non-backshifting as it has no Past (Past, as noted in Section 4.5, would enforce temporal backshift). (42b) and (42c) are backshifting, as suggested by Past in the consequent. This classification is supported by the fact that only (42b) and (42c) can involve past adverbials in the antecedent, a hallmark of backshifting CFs (see Section 4.5).

- (44) a. [John-ga **kinoo** ku]-reba, [paatii-wa motto moriagar]-u
 John-NOM yesterday come-COND party-TOP more be.fun-NPST
kamosirena-i (noni).
 might-NPST SFP
 Lit. ‘If John came yesterday, the party might be more fun.’ (#: 4/4)
- b. [John-ga **kinoo** ku]-reba, [paatii-wa motto moriagar]-ta
 John-NOM yesterday come-COND party-TOP more be.fun-PAST
kamosirena-i (noni).
 might-NPST SFP
 ‘If John had come yesterday, the party might have been more fun.’ (ok: 4/4)
- c. [John-ga **kinoo** ku]-reba, [paatii-wa motto moriagar]-u
 John-NOM yesterday come-COND party-TOP more be.fun-NPST
kamosirena-katta (noni).
 might-PAST SFP
 ‘If John had come yesterday, the party might have been more fun.’ (ok: 4/4)

What structures and interpretations will each form have? To discuss (42b) first, I predict that it maps uniquely to backshifting CFs with the Wide-Scope structure.³²

$$(45) \quad \phi\text{-COND}, \psi\text{-PAST } kamosirena\text{-NPST} \rightsquigarrow [\Diamond^R [\text{PAST} [\Box_{\text{IF-}\phi}^* [\text{NPST } \psi]]]]$$

This prediction should be straightforward: given that Past in the consequent enforces temporal backshift, the modal restricted by the antecedent has to be in the scope of Past, so that its domain expand through temporal backshift; however, as inferred from the surface order, *kamosirena*- here is structurally higher than Past (see Section 3.3); this indicates that it is not what is restricted by the antecedent but there is a covert necessity modal inserted below Past and targeted by the antecedent; the possibility modal, then, outscopes an entire (backshifting) *would*-CF, thus instantiating the idea of Wide-Scope.

For (42c), I predict two possibilities: it may map to backshifting CFs with either the One-Modal structure or the Narrow-Scope structure.

$$(46) \quad \phi\text{-COND}, \psi\text{-NPST } kamosirena\text{-PAST} \rightsquigarrow$$

a. [PAST [$\Diamond_{\text{IF-}\phi}^*$ [NPST ψ]]] (One-Modal)

³²As noted in Section 4.5, I assume that the modal restricted by a CF antecedent is construed metaphorically, as indicated by $\approx$. Recall also that overt Non-Past on the topmost modal (here the possibility modal) is abbreviated for simplification purposes (see Section 4.3), and that a covert Non-Past is inserted in the consequent so that forward-shifting of ψ can be accounted for in cases like this (see Section 4.5). The flavor of the possibility modal is left open, as indicated by R .

$$\text{b. } [\text{PAST } [\Box_{\text{IF}-\phi}^* [\Diamond_{\text{IF}-\phi}^R [\text{NPST } \psi]]]] \quad (\text{Narrow-Scope})$$

As inferred from the surface order, *kamosirena*- here is structurally lower than Past. Temporal backshift may thus directly operate on it, resulting in the structure of One-Modal (46a). But it is also possible for a covert necessity modal to be inserted below Past, restricted by the antecedent and targeted for temporal backshift: the possibility modal then takes narrow scope, hence the structure of Narrow-Scope (46b).³³

Finally, for (42a), I predict three possibilities: it may map to non-backshifting CFs with the One-Modal structure, the Wide-Scope structure, or the Narrow-Scope structure.

$$\begin{aligned} (47) \quad & \phi\text{-COND}, \psi\text{-NPST } kamosirena\text{-NPST } \rightsquigarrow \\ & \text{a. } [\Diamond_{\text{IF}-\phi}^* [\text{NPST } \psi]] \quad (\text{One-Modal}) \\ & \text{b. } [\Diamond^R [\Box_{\text{IF}-\phi}^* [\text{NPST } \psi]]] \quad (\text{Wide-Scope}) \\ & \text{c. } [\Box_{\text{IF}-\phi}^* [\Diamond_{\text{IF}-\phi}^R [\text{NPST } \psi]]] \quad (\text{Narrow-Scope}) \end{aligned}$$

Table 1 summarizes the overall prediction. The predicted form-meaning mapping is further ramified by specifying the flavor of the possibility modal left open (i.e., $\Diamond^R$). I henceforth use “ $\phi \Diamond \rightsquigarrow_{\text{NBS}} \psi$ ”, and so on, as a mnemonic device for their associated linguistic structures and interpretations (“NBS” and “BS” for “non-backshifting” and “backshifting” respectively). The gray-colored ones in the table, namely, $\Diamond^{\sim}(\phi \triangleright_{\text{NBS}}^{\sim} \psi)$ and $\Diamond^{\sim}(\phi \triangleright_{\text{BS}}^{\sim} \psi)$, are excluded from consideration, as they are equivalent to their respective embedded *would*-CFs (see Appendix B for formal proofs).

In the following sections, I will examine this prediction against empirical data. As a disclaimer, I leave open whether $\Diamond^{\sim}(\phi \triangleright_{\text{NBS}}^{\sim} \psi)$ is attested in Japanese. As the table shows, non-backshifting *might*-CFs could be interpreted in many ways, and I currently have no diagnostic that clearly discriminates this from other construals. Finding a good diagnostic is left for future work.

Other predictions, however, are all testable. I will first examine the two types of backshifting *might*-CFs (Section 6). I will then turn to One-Modal and Narrow-Scope (Section 7). I will finally discuss $\phi \triangleright_{\text{NBS}}^{\sim} \Diamond^{\sim} \psi$ and $\phi \triangleright_{\text{BS}}^{\sim} \Diamond^{\sim} \psi$, namely, Narrow-Scope with epistemic possibility modals (Section 8).

6 Testing predictions (i): teasing apart the two backshifting *might*-CFs

In this section, I demonstrate that the two sentential forms, “ $\phi\text{-COND}, \psi\text{-PAST } kamosirena\text{-NPST}$ ” and “ $\phi\text{-COND}, \psi\text{-NPST } kamosirena\text{-PAST}$ ”, involve different structures and interpretations of *might*-CFs.

³³Recall that the possibility modal here has to inherit the antecedent restriction from the necessity modal (see Section 4.4).

ϕ -COND, ψ -NPST <i>kamosirena</i> -NPST	$\llbracket \Diamond_{\text{IF-}\phi}^* \llbracket \text{NPST } \psi \rrbracket \rrbracket$	One-Modal $\phi \Diamond \rightarrow_{\text{NBS}}^{\sim} \psi$
	$\llbracket \Diamond \sim \llbracket \Box_{\text{IF-}\phi}^* \llbracket \text{NPST } \psi \rrbracket \rrbracket \rrbracket$	Wide-Scope $\Diamond^{\sim}(\phi \triangleright_{\text{NBS}}^{\sim} \psi)$
	$\llbracket \Diamond \sim \llbracket \Box_{\text{IF-}\phi}^* \llbracket \text{NPST } \psi \rrbracket \rrbracket \rrbracket$	Wide-Scope $\Diamond^{\sim}(\phi \triangleright_{\text{NBS}}^{\sim} \psi)$
	$\llbracket \Box_{\text{IF-}\phi}^* \llbracket \Diamond_{\text{IF-}\phi}^* \llbracket \text{NPST } \psi \rrbracket \rrbracket \rrbracket$	Narrow-Scope $\phi \triangleright_{\text{NBS}}^{\sim} \Diamond^{\sim} \psi$
	$\llbracket \Box_{\text{IF-}\phi}^* \llbracket \Diamond_{\text{IF-}\phi}^* \llbracket \text{NPST } \psi \rrbracket \rrbracket \rrbracket$	Narrow-Scope $\phi \triangleright_{\text{NBS}}^{\sim} \Diamond^{\sim} \psi$
ϕ -COND, ψ -PAST <i>kamosirena</i> -NPST	$\llbracket \Diamond \sim \llbracket \text{PAST} \llbracket \Box_{\text{IF-}\phi}^* \llbracket \text{NPST } \psi \rrbracket \rrbracket \rrbracket \rrbracket$	Wide-Scope $\Diamond^{\sim}(\phi \triangleright_{\text{BS}}^{\sim} \psi)$
	$\llbracket \Diamond \sim \llbracket \text{PAST} \llbracket \Box_{\text{IF-}\phi}^* \llbracket \text{NPST } \psi \rrbracket \rrbracket \rrbracket \rrbracket$	Wide-Scope $\Diamond^{\sim}(\phi \triangleright_{\text{BS}}^{\sim} \psi)$
ϕ -COND, ψ -NPST <i>kamosirena</i> -PAST	$\llbracket \text{PAST} \llbracket \Diamond_{\text{IF-}\phi}^* \llbracket \text{NPST } \psi \rrbracket \rrbracket \rrbracket$	One-Modal $\phi \Diamond \rightarrow_{\text{BS}}^{\sim} \psi$
	$\llbracket \text{PAST} \llbracket \Box_{\text{IF-}\phi}^* \llbracket \Diamond_{\text{IF-}\phi}^* \llbracket \text{NPST } \psi \rrbracket \rrbracket \rrbracket \rrbracket$	Narrow-Scope $\phi \triangleright_{\text{BS}}^{\sim} \Diamond^{\sim} \psi$
	$\llbracket \text{PAST} \llbracket \Box_{\text{IF-}\phi}^* \llbracket \Diamond_{\text{IF-}\phi}^* \llbracket \text{NPST } \psi \rrbracket \rrbracket \rrbracket \rrbracket$	Narrow-Scope $\phi \triangleright_{\text{BS}}^{\sim} \Diamond^{\sim} \psi$

Table 1 Summary of the predicted mapping from forms to structures/interpretations.

6.1 Singling out Wide-Scope

Recall the prediction from Section 5: the form “ ϕ -COND, ψ -PAST *kamosirena*-NPST” is predicted to correspond uniquely to Wide-Scope (48), whereas the form “ ϕ -COND, ψ -NPST *kamosirena*-PAST” may map to One-Modal (49a) or Narrow-Scope (49b). The difference between One-Modal and Narrow-Scope does not concern us here.³⁴

- (48) ϕ -COND, ψ -PAST *kamosirena*-NPST:
 $\llbracket \Diamond \sim \llbracket \text{PAST} \llbracket \Box_{\text{IF-}\phi}^* \llbracket \text{NPST } \psi \rrbracket \rrbracket \rrbracket \rrbracket$
 $= \lambda i. \exists j \in \mathcal{D}^{\sim}(i). \exists k < j. \forall l \in \mathcal{D}^{\sim*}(k) \cap \llbracket \phi \rrbracket. \exists m \geq l. \llbracket \psi \rrbracket(m)$
- (49) ϕ -COND, ψ -NPST *kamosirena*-PAST:
a. $\llbracket \text{PAST} \llbracket \Diamond_{\text{IF-}\phi}^* \llbracket \text{NPST } \psi \rrbracket \rrbracket \rrbracket$
 $= \lambda i. \exists j < i. \exists k \in \mathcal{D}^{\sim*}(j) \cap \llbracket \phi \rrbracket. \exists l \geq k. \llbracket \psi \rrbracket(l)$
b. $\llbracket \text{PAST} \llbracket \Box_{\text{IF-}\phi}^* \llbracket \Diamond_{\text{IF-}\phi}^* \llbracket \text{NPST } \psi \rrbracket \rrbracket \rrbracket \rrbracket$
 $= \lambda i. \exists j < i. \forall k \in \mathcal{D}^{\sim*}(j) \cap \llbracket \phi \rrbracket. \exists l \in \mathcal{D}^{\sim}(k) \cap \llbracket \phi \rrbracket. \exists m \geq l. \llbracket \psi \rrbracket(m)$

From a truth-conditional standpoint, (48) and (49a)/(49b) express different types of meaning. (48) concerns the speaker’s *epistemic state*: it states that it is epistemically conceivable that there was a time at which all the ϕ -indices in the domain were also ψ -indices. On the other hand, (49a) and (49b) concern *metaphysical facts*: (49a) states that there was a time at which some ϕ -index in the domain was also a ψ -index; (49b) states that there was a time at which all the indices in the domain were also $\Diamond\psi$ -indices.

This semantic difference between (48) and (49a)/(49b) leads to the following prediction: (48) and (49a)/(49b) require different information for the speaker to *assert* them. Assuming that one can assert a proposition only when one believes it, assertion of (49a)/(49b) requires the speaker to believe that there was actually a moment at which some ϕ -index in the domain was a ψ -index, or all the ϕ -indices in the domain were $\Diamond\psi$ -indices. If the speaker is uncertain whether there existed such a moment, they

³⁴Here I ignore $\phi \triangleright_{\text{BS}}^{\sim} \Diamond^{\sim} \psi$ because it involves additional complications, whose discussion I defer until Section 8.

are unassertable. In contrast, assertion of (48) requires that it be compatible with the speaker’s epistemic state that there was a time at which all the ϕ -indices in the domain were ψ -indices: one can assert it even when one is completely ignorant of the relevant matter, because in such cases everything is compatible with one’s epistemic state.

The prediction is thus the following: in contexts that force the speaker’s complete ignorance concerning the situation under discussion, only (48) is assertable; (49a) and (49b) are unassertable (thus infelicitous) because in such contexts the speaker would be uncertain whether there actually was a moment at which some ϕ -index was a ψ -index, or all the ϕ -indices were $\Diamond\psi$ -indices.

This prediction is borne out. In the context of (50), the speaker is completely uninformed about the restaurant that she did not enter.³⁵ All the four Japanese speakers found a contrast: only ϕ -COND, ψ -PAST *kamosirena*-NPST can be felicitously used.³⁶

- (50) **Amex:** You are traveling in Japan. You found two restaurants standing alongside one another, Restaurant A and Restaurant B, and entered Restaurant A. After you ordered food, you came to wonder if they accept American Express. You asked a staff member, and he told you that they don’t. This was disappointing because the only card you have is Amex. While you have no information about Restaurant B, you imagine:

- a. *B-ni hair-eba, Amex-ga tuka-e-ta kamosirena-i.*
B-DAT enter-COND Amex-NOM use-able-PAST might-NPST
‘If I had entered B, I might have been able to use Amex.’ (ok: 4/4)
- b. *B-ni hair-eba, Amex-ga tuka-e-ru kamosirena-katta.*
B-DAT enter-COND Amex-NOM use-able-NPST might-PAST
‘If I had entered B, I might have been able to use Amex.’ (? : 1/4, ?? : 1/4, #: 2/4)

(51) illustrates the same point from a different perspective. Suppose that we are engaging in a thought experiment concerning the inventor Thomas Edison: what reaction would he have shown if he had seen a smartphone while he was alive? Possibility 1: It could be that he would have been utterly astonished; this is plausible given that the

³⁵The context here can be considered a variant of DeRose’s scenario from Section 2, but it is more radical in that it forces complete ignorance of the past $\phi \wedge \psi$ -possibilities, which is not clearly controlled for in DeRose’s context. I also note that the English speakers I consulted found the English translation of (50) (‘If I had entered B, I might have been able to use Amex.’) felicitous in this context.

³⁶For comparison, the same sentential form as (50b) is felicitous in the context of (i). Unlike the context for (50b), the context here implies the existence of a past moment that makes One-Modal and Narrow-Scope assertable.

- (i) (There are two restaurants in town. Restaurant W does not accept Amex. Restaurant X accepts Amex, but its register is broken in a strange way; every time one tries Amex, there is a 50% chance of failure, and when that happens is at random. The speaker is familiar with all this, but today he happens to have left all his cards except Amex at home. Finding there is little cash in the wallet, he decides to go straight home. After returning home, he says to his spouse:)
- [*Kyoo-wa massugu kaet-te ki-ta yo.*] *Tasikani, X-ni hair-eba, Amex-ga*
today-TOP straight go.home-GRND come-PAST SFP sure X-DAT enter-COND Amex-NOM
tuka-e-ru kamosirena-katta [*kedo, dame dat-ta toki-ni komar-u daroo?*]
use-able-NPST might-PAST but useless COP-PAST when-DAT be.in.trouble-NPST MODAL
‘[I came home straight today.] Sure, if I had entered X, I might have been able to use Amex[, but I would be in trouble if it didn’t work, you know.]’ (ok: 4/4)

technologies used for smartphones would have been beyond comprehension of those who lived almost a century ago. Possibility 2: It could be that he would *not* have been surprised at all; this may not be as plausible as Possibility 1, but we cannot say that it is impossible; no one other than him could have known the answer. Now notice that this context licenses the use of $\Diamond \sim (\phi \triangleright_{BS}^{\sim} \psi)$: since this is a thought experiment whose answer we will never know, no one can deny Possibility 1 (i.e., the possibility of $\phi \triangleright_{BS}^{\sim} \psi$), thus making $\Diamond \sim (\phi \triangleright_{BS}^{\sim} \psi)$ true and assertable. Meanwhile, it does *not* license the use of $\phi \Diamond \rightarrow_{BS}^{\sim} \psi$ or $\phi \triangleright_{BS}^{\sim} \Diamond \sim \psi$. Possibility 2 implies the epistemic possibility of $\phi \triangleright_{BS}^{\sim} \neg \psi$, whose truth conditions are incompatible with those of both $\phi \Diamond \rightarrow_{BS}^{\sim} \psi$ and $\phi \triangleright_{BS}^{\sim} \Diamond \sim \psi$. Because Possibility 2 is no less unrefutable, the speaker must believe that $\phi \triangleright_{BS}^{\sim} \neg \psi$ is epistemically possible. But this then prevents the speaker from being certain about the truth of $\phi \Diamond \rightarrow_{BS}^{\sim} \psi$ and $\phi \triangleright_{BS}^{\sim} \Diamond \sim \psi$, hence rendering them unassertable. This prediction is confirmed: again, only ψ -PAST *kamosirena*-NPST is felicitous.³⁷

- (51) a. *Mosi Edison-ga sumaho-o mir-eba, sono gizyutu-no*
 ADV Edison-NOM smartphone-ACC see-COND that technology-GEN
takasa-ni hidoku odoroi-ta kamosirena-i.
 advance-DAT terribly be.surprised-PAST might-NPST
 Lit. ‘If Edison had seen a smartphone, he might have been terribly surprised at its technology.’ (ok: 3/4, ??: 1/4)
- b. *Mosi Edison-ga sumaho-o mir-eba, sono gizyutu-no*
 ADV Edison-NOM smartphone-ACC see-COND that technology-GEN
takasa-ni hidoku odorok-u kamosirena-katta.
 advance-DAT terribly be.surprised-NPST might-PAST
 Lit. ‘If Edison had seen a smartphone, he might have been terribly surprised at its technology.’ (#: 4/4)

These data suggest that the two varieties of backshifting *might*-CFs in Japanese do involve different structures and interpretations. I will show further evidence below.

6.2 Singling out One-Modal/Narrow-Scope

When is $\Diamond \sim (\phi \triangleright_{BS}^{\sim} \psi)$ infelicitous while $\phi \Diamond \rightarrow_{BS}^{\sim} \psi$ and $\phi \triangleright_{BS}^{\sim} \Diamond \sim \psi$ are not? A straightforward prediction is that it is infelicitous when $\phi \triangleright_{BS}^{\sim} \psi$ (i.e., the *would*-CF it embeds) is known to be false, and in the same context, $\phi \Diamond \rightarrow_{BS}^{\sim} \psi$ and $\phi \triangleright_{BS}^{\sim} \Diamond \sim \psi$ may not be. After all, if one knows that $\phi \triangleright_{BS}^{\sim} \psi$ is false, that means that it is incompatible

³⁷While finding a contrast here, the speaker who gave “??” to (51a) reported that the non-backshifting form sounds better: the speaker reported that (51a) tends to trigger the wrong presupposition that smartphones did exist in Edison’s time (the speaker said that the noun “smartphone” feels “undefined” in (51a)), though the other three speakers did not find such an implication in (51a). It is worth noting that the English translation here, especially the use of Past Perfect in the antecedent, seems to trigger the same presupposition as the Japanese consultant reported (I thank Si Kai Lee for pointing this out; he suggested ‘if Edison *were to have seen* a smartphone, he might have come to express surprise at the level of technology’ as a more natural paraphrase of the English sentence here).

A reviewer of *L&P* suggested that *-ba* might not be a perfect fit here. Although the effect of the choice of a connective is beyond the scope of this paper, those who have difficulty with *-ba* are invited to use another connective “-tara”. I checked the acceptability of (51a)/(51b) with *-tara*, and the distribution was almost the same ((51a): ok: 3/4, #: 1/4; (51b): ??: 1/4, #: 3/4).

with one’s epistemic state, thus making $\Diamond\sim(\phi \triangleright_{\text{BS}}^{\sim} \psi)$ false accordingly. The falsity of $\phi \triangleright_{\text{BS}}^{\sim} \psi$, however, is not incompatible with the truth of $\phi \Diamond\rightarrow_{\text{BS}}^{\sim} \psi$ and $\phi \triangleright_{\text{BS}}^{\sim} \Diamond\sim\psi$.

Consider (52). Since the winner is selected by a random process, $\phi \triangleright_{\text{BS}}^{\sim} \psi$ (i.e., *if I had answered the questionnaire, I would have won a gift certificate*) is false. Thus, the form “ ϕ -COND, ψ -PAST *kamosirena*-NPST”, which I predict to express $\Diamond\sim(\phi \triangleright_{\text{BS}}^{\sim} \psi)$, will be false too. Meanwhile, the form “ ϕ -COND, ψ -NPST *kamosirena*-PAST”, which I predict to express $\phi \Diamond\rightarrow_{\text{BS}}^{\sim} \psi$ or $\phi \triangleright_{\text{BS}}^{\sim} \Diamond\sim\psi$, will be true, because it was metaphysically possible for any respondent to win a gift certificate.

- (52) **Questionnaire:** The university invited the students to answer a questionnaire that offered an expensive gift certificate to *one randomly selected respondent*. John received the questionnaire, but he did not notice that there would be a lottery and ended up not answering it. As it happened, he became aware of the lottery only after he learned that his roommate happened to have won it. The following is John’s reaction upon discovering all this:

- a. *[Mazi ka yo,] ano ankeeto-ni kotaer-eba, gifutoken-ga*
 serious Q SFP that questionnaire-DAT answer-COND gift.certificate-NOM
atat-ta kamosirena-i no ka! [Sippaisi-ta yo.]
 get.won-PAST might-NPST FIN Q fail-PAST SFP
 ‘[Shoot,] if I had answered that questionnaire, I might have won a gift certificate! [What a mistake that was.]’ (ok: 1/4, ?: 1/4, ??: 1/4, #: 1/4)
- b. *[Mazi ka yo,] ano ankeeto-ni kotaer-eba, gifutoken-ga*
 serious Q SFP that questionnaire-DAT answer-COND gift.certificate-NOM
atar-u kamosirena-katta no ka! [Sippaisi-ta yo.]
 get.won-NPST might-PAST FIN Q fail-PAST SFP
 ‘[Shoot,] if I had answered that questionnaire, I might have won a gift certificate! [What a mistake that was.]’ (ok: 4/4)

Insofar as the reported judgments go, there seems to be speaker variation concerning the acceptability of (52a): while three found a contrast to (52b), one gave “ok” to both (52a) and (52b). The comments from the latter speaker, however, actually point to the semantic difference we predict. Specifically, the speaker who gave “ok” to (52a) reported that they would imagine the utterer of (52a) to be a careless person who easily forgets the rule of the lottery and hastily concludes he would have won if only he had answered the questionnaire; if the utterer of (52a) is known not to be careless, they would imagine this person to be joking, pretending to not understand the rule of the lottery. Although the sentence somehow struck them as acceptable, these comments suggest that (52a) seems to instantiate $\Diamond\sim(\phi \triangleright_{\text{BS}}^{\sim} \psi)$ in this speaker’s intuition too. Note also that the overall distribution of judgments in any case implies the contrast between (52a) and (52b), though not as clearcut as those from the last section.^{38,39}

³⁸I do not fully understand why (52a) does not sound very bad to some speakers. One possibility I can think of is that, although $\phi \triangleright_{\text{BS}}^{\sim} \psi$ and $\Diamond\sim(\phi \triangleright_{\text{BS}}^{\sim} \psi)$ are equally false in this context, the presence of the possibility modal gives a sort of hedging nuance to the latter, thus rendering the badness less stark than that of the former. I also note that the English speakers I consulted found the English translation of (52) (‘if I had answered that questionnaire, I might have won a gift certificate’) felicitous in the given context.

³⁹A reviewer for *L&P* draws my attention to corresponding data from German, another language that can express scope relations between modal and temporal ingredients transparently. In German, relevant

To show more clearly that the two forms of *might*-CFs semantically come apart, let me show another piece of evidence. The diagnostic I use is the scope interaction with nominal quantifiers, the basics of which I already introduced in Section 3.4. To recap, epistemic modals must outscope nominal quantifiers in the same clause, according to the Epistemic Containment Principle (von Fintel and Iatridou 2003). As was shown in Section 3.4, Japanese *kamosirena*- is not an exception to this constraint.

Applying this to the *might*-CFs at hand, it is predicted that when a nominal quantifier is involved in the consequent of a *might*-CF, the quantifier cannot outscope *kamosirena*- if the latter is interpreted epistemically.

A more specific prediction is the following: (i) ϕ -COND, ψ -NPST *kamosirena*-PAST, which I predict to express $\phi \Diamond \rightarrow_{BS} \psi$ / $\phi \triangleright_{BS} \Diamond \approx \psi$, will allow a nominal quantifier in the consequent to outscope the entire sentence, because there are no epistemically construed modals that would block highest scope of the nominal quantifier, but (ii) ϕ -COND, ψ -PAST *kamosirena*-NPST, which I predict to express $\Diamond \sim (\phi \triangleright_{BS} \psi)$, will not, because the topmost $\Diamond \sim$ would block it. (53) summarizes this prediction.

- (53) a. ϕ -COND, ψ -NPST *kamosirena*-PAST: $\checkmark$ NOM-Q $>$ $\phi \Diamond \rightarrow_{BS} \psi$, $\phi \triangleright_{BS} \Diamond \approx \psi$
b. ϕ -COND, ψ -PAST *kamosirena*-NPST: * NOM-Q $>$ $\Diamond \sim (\phi \triangleright_{BS} \psi)$

To first examine the prediction (53a), the sentence in (54) below involves a nominal universal quantifier in the consequent. The context forces interpretations on which the nominal quantifier outscoops the entire conditional (i.e., ‘for every student x , if x had answered the questionnaire, x might have won a gift certificate’).⁴⁰ All four speakers found (54) acceptable, thus confirming the prediction (53a).⁴¹

scope interactions take place between a modal and the Perfect marker “*haben*” (‘have’): the scope relations between them are overtly encoded in such a way that the one that appears in the V2 position always outscoops the one that appears in the infinitival position. The reviewer points out that the German equivalents of (52a) and (52b), which are shown in (i), exhibit a contrast similar to what we predict for Japanese. (ia) is a structure in which the possibility modal outscoops Perfect, hence expected to correspond to Japanese (52a) (i.e., expected to represent $\Diamond \sim (\phi \triangleright_{BS} \psi)$), whereas (ib) is a structure in which Perfect outscoops the possibility modal, hence expected to correspond to Japanese (52b) (i.e., expected to represent $\phi \Diamond \rightarrow_{BS} \psi$ / $\phi \triangleright_{BS} \Diamond \approx \psi$). According to the reviewer, (ia) is fairly degraded in a similar context, while (ib) is not (I confirmed this judgment with another German speaker).

- (i) (John didn’t answer the questionnaire, but he realized afterwards that one randomly chosen respondent would receive a gift certificate. John thinks:)
a. ?? *Wenn ich den Fragebogen ausgefüllt hätte, könnte ich den Gutschein gewonnen haben.*
if I the questionnaire filled.in have.SUBJ can.SUBJ I the gift.certificate win have
‘If I had answered the questionnaire, I might have won a gift certificate.’
b. *Wenn ich den Fragebogen ausgefüllt hätte, hätte ich den Gutschein gewinnen können.*
if I the questionnaire filled.in have.SUBJ have.SUBJ I the gift.certificate win can
‘If I had answered the questionnaire, I might have won a gift certificate.’

The contrast reported by the reviewer lends indirect support to my account of Japanese *might*-CFs, but a larger speaker sample will be needed to figure out how robust the purported contrast is in German. Further research on German *might*-CFs is left for another occasion.

⁴⁰The other scope possibility is the quantifier scoping *within* the consequent, but then the sentence would have a strange construal (i.e., ‘if someone had answered the questionnaire, every student might have won a gift certificate’, with the null subject in the antecedent filled by a specific individual), which conflicts with the stipulation of the context that there was only one gift certificate to be offered.

⁴¹I note that the English speakers I consulted found the English translation of (54) *infelicitous*, reporting that it is difficult to get readings on which *every* scopes over the entire conditional (this was also the case

- (54) **Questionnaire** ($\forall$): The university invited the students to answer a questionnaire which would offer an expensive gift certificate to *one randomly selected respondent*. Contrary to the university staff’s expectations, no student ended up answering the questionnaire. Faced with this result, a staff member says to her colleague:

[Konna tyansu minogas-u nante mottaina-i yo ne.] Kotae-sae
 such chance miss-NPST that be.wasteful-NPST SFP SFP answer-just
 su-reba, dono gakusee-ni-mo gifutoken-ga atar-u
 do-COND which student-DAT-also gift.certificate-NOM get-NPST
 kamosirena-**katta** noni.
 might-PAST SFP

Lit. ‘[It’s a shame that they missed a chance like this.] Every student might have won a gift certificate if they had just answered it.’ (ok: 4/4)

Turning to the prediction (53b), (55) involves the sentence of the form “ ϕ -COND, ψ -PAST *kamosirena*-NPST”, which I predict to express $\Diamond^{\sim}(\phi \triangleright_{\text{BS}}^{\sim} \psi)$. The possibility of the nominal quantifier scoping below $\Diamond^{\sim}$ is precluded by the contextual stipulation that only one student could win the lottery.⁴² Crucially, all the speakers found (55) significantly or extremely degraded. The predictions in (53) are thus confirmed.

- (55) **Questionnaire** ($\forall$): (the same context as in (54))

[Konna tyansu minogas-u nante mottaina-i yo ne.] Kotae-sae
 such chance miss-NPST that be.wasteful-NPST SFP SFP answer-just
 su-reba, dono gakusee-ni-mo gifutoken-ga atat-**ta** kamosirena-**i**
 do-COND which student-DAT-also gift.certificate-NOM get-PAST might-NPST
 noni.
 SFP

Lit. ‘[It’s a shame that they missed a chance like this.] Every student might have won a gift certificate if they had just answered it.’ (?: 1/4, #: 3/4)

The result is replicated with additive quantifiers. (56) involves the form “ ϕ -COND, ψ -NPST *kamosirena*-PAST”. Here again, the context forces readings on which the nominal quantifier outscopes the entire conditional (i.e., the truth of the *might*-CF applies to John as well as others). Most of the speakers found (56) acceptable.

- (56) **Questionnaire** (**too**): The university invited all the students—both undergrad and grad—to answer a questionnaire that offered a gift certificate to *one randomly selected respondent*. John, a grad student, received the questionnaire, but he mistakenly thought it was only for undergrads and ended up ignoring it. He only became aware that he could have answered it after he

with the English translation of (56), which involves *too*). This seems to suggest that *might* in this English *might*-CF is construed epistemically, thus apparently supporting the analysis of Wide-Scope, but it is not fully clear why it has to be epistemic in this particular case. I have to leave this issue for future work.

⁴²If the scope relation were $\Diamond^{\sim} > \forall > (\phi \triangleright_{\text{BS}}^{\sim} \psi)$, the sentence would mean something like “it is epistemically possible that every student would have won a gift certificate if they had answered the questionnaire”, which cannot be true given the rule of the lottery. If $\forall$ scoped within the consequent, the sentence, like in (54), would have a strange construal (i.e., ‘it is possible that if someone had answered the questionnaire, every student would have won a gift certificate’), which is again at odds with the present context.

learned that his colleague had won the lottery and received the gift certificate. The following is what the colleague says to someone:

[*John-wa mottainai koto-o si-ta yo.*] *Kotae-sae su-reba,*
 John-TOP wasteful thing-ACC do-PAST SFP answer-just do-COND
kare-ni-mo gifutoken-ga atar-u kamosirena-katta noni.
 he-DAT-also gift.certificate-NOM get.won-NPST might-PAST SFP
 Lit. ‘[John missed such a good chance.] If he had answered the questionnaire, he, too, might have won a gift certificate.’ (ok: 3/4, ?: 1/4)

(57) involves the form “ ϕ -COND, ψ -PAST *kamosirena*-NPST”. Like above, the context precludes the possibility of *too* scoping below $\Diamond^{\sim}$ of $\Diamond^{\sim}(\phi \triangleright_{BS}^{\sim} \psi)$. All the speakers found the sentence significantly or extremely degraded.⁴³ The predictions in (53) are thus borne out with additive quantifiers too.

(57) **Questionnaire (too):** (the same context as in (56))

[*John-wa mottainai koto-o si-ta yo.*] *Kotae-sae su-reba,*
 John-TOP wasteful thing-ACC do-PAST SFP answer-just do-COND
kare-ni-mo gifutoken-ga atat-ta kamosirena-i noni.
 he-DAT-also gift.certificate-NOM get.won-PAST might-NPST SFP
 Lit. ‘[John missed such a good chance.] If he had answered the questionnaire, he, too, might have won a gift certificate.’ (?: 2/4, #: 2/4)

The two forms of backshifting *might*-CFs in Japanese thus involve different structures and interpretations. See also Appendix D, where I discuss the potential influence of the presence of an aspectual marker on the acceptability of each sentence.

7 Testing predictions (ii): One-Modal vs. Narrow-Scope

I turn to differentiating between One-Modal and Narrow-Scope, which, as philosophers have noted, seem to have a similar range of interpretation. One-Modal, namely $\phi \Diamond \rightarrow_{NBS}^{\sim} \psi$ in (58) and $\phi \Diamond \rightarrow_{BS}^{\sim} \psi$ in (59), involve structures in which a metaphysically construed $\Diamond$ is directly restricted by a CF antecedent. (58) and (59) involve existential quantification over a forward-extended domain, requiring that the domain involve at least one index at which the antecedent and consequent are both true.

$$(58) \quad \llbracket \Diamond_{IF-\phi}^{\sim+} [NPST \psi] \rrbracket \\ = \lambda i. \exists j \in \mathcal{D}_+^{\sim*}(i) \cap \llbracket \phi \rrbracket. \exists k \geq j. \llbracket \psi \rrbracket(k)$$

$$(59) \quad \llbracket PAST [\Diamond_{IF-\phi}^{\sim*} [NPST \psi]] \rrbracket \\ = \lambda i. \exists j < i. \exists k \in \mathcal{D}^{\sim*}(j) \cap \llbracket \phi \rrbracket. \exists l \geq k. \llbracket \psi \rrbracket(l)$$

Narrow-Scope, namely $\phi \triangleright_{NBS}^{\sim} \Diamond^{\sim} \psi$ in (60) and $\phi \triangleright_{BS}^{\sim} \Diamond^{\sim} \psi$ in (61), involve structures in which a metaphysically construed $\Diamond$ appears in the consequent of a *would*-CF.

⁴³As predicted, the speakers reported that the only reading of (57) is ‘it is possible that John would have won a gift certificate if he had answered the questionnaire, in addition to the one who actually won it’.

(60) and (61) involve *point-wise* evaluation of $\Diamond$, requiring that for every index in the domain at which the antecedent is true, there be at least one co-temporal index which is metaphysically accessible from it and at which the consequent is also true.

$$(60) \quad \begin{aligned} & \llbracket \Box_{\text{IF-}\phi}^+ [\Diamond_{\text{IF-}\phi} [\text{NPST } \psi]] \rrbracket \\ & = \lambda i. \forall j \in \mathcal{D}_+^*(i) \cap \llbracket \phi \rrbracket. \exists k \in \mathcal{D}^\sim(j) \cap \llbracket \phi \rrbracket. \exists l \geq k. \llbracket \psi \rrbracket(l) \end{aligned}$$

$$(61) \quad \begin{aligned} & \llbracket \text{PAST} [\Box_{\text{IF-}\phi}^* [\Diamond_{\text{IF-}\phi} [\text{NPST } \psi]]] \rrbracket \\ & = \lambda i. \exists j < i. \forall k \in \mathcal{D}^*(j) \cap \llbracket \phi \rrbracket. \exists l \in \mathcal{D}^\sim(k) \cap \llbracket \phi \rrbracket. \exists m \geq l. \llbracket \psi \rrbracket(m) \end{aligned}$$

While they cover a similar range of interpretation, One-Modal and Narrow-Scope are not semantically equivalent, at least in the present formalization. More specifically, $\phi \triangleright_{\text{NBS}}^\sim \Diamond^\sim \psi$ and $\phi \triangleright_{\text{BS}}^\sim \Diamond^\sim \psi$ are truth-conditionally *stronger* than $\phi \Diamond \rightarrow_{\text{NBS}}^\sim \psi$ and $\phi \Diamond \rightarrow_{\text{BS}}^\sim \psi$, respectively. Why this is so will be clarified shortly with concrete examples; for formal proofs, I refer the reader to Appendix B.

Given that Narrow-Scope asymmetrically entails One-Modal, there has to be a sentence that is false under the former but true under the latter: One-Modal will thus have to be adopted or rejected depending on whether native speakers judge the sentence to be true or false, respectively. The testing cases here are the forms “ ϕ -COND, ψ -NPST *kamosirena*-NPST” (non-backshifting) and “ ϕ -COND, ψ -NPST *kamosirena*-PAST” (backshifting), both of which are predicted to allow for the two interpretations in question. I first discuss the backshifting form and then turn to the non-backshifting form.

Recall the truth conditions. The existence of a $\phi \wedge \psi$ -index in the forward-extended domain suffices for the truth of $\phi \Diamond \rightarrow_{\text{BS}}^\sim \psi$ but not for $\phi \triangleright_{\text{BS}}^\sim \Diamond^\sim \psi$: if there is a ϕ -index in the domain at which $\neg \Diamond \psi$ holds, the sentence is false with $\phi \triangleright_{\text{BS}}^\sim \Diamond^\sim \psi$.

A situation that falsifies $\phi \triangleright_{\text{BS}}^\sim \Diamond^\sim \psi$ can be created by exploiting the features of forward-extension. As shown in Section 4.4, forward-extension of the domain enables access to future indices. However, historical alternatives may depart from each other in the future, as the $T \times W$ -frame dictates.⁴⁴ Because of this, the forward-extended domain restricted by the antecedent “ ϕ ” may involve two distinct classes of historical alternatives, namely, one that includes at least one ψ -index, and one that includes no ψ -index. In such cases, only $\phi \Diamond \rightarrow_{\text{BS}}^\sim \psi$ is true: it is true because the domain involves a $\phi \wedge \psi$ -index; $\phi \triangleright_{\text{BS}}^\sim \Diamond^\sim \psi$ is false because $\neg \Diamond \psi$ holds unanimously in the latter class of historical alternatives.

One concrete context that instantiates such a situation is (62), which involves two types of coin.

- (62) **Two coins (backshifting):** The following challenge was offered to John: “There are two coins, a fair one and a double-tailed one. If you take up the challenge, you first pick one coin up randomly, and then toss it. If it has landed heads, you win a prize. If it has landed tails, you lose money.” After much contemplation, John chose not to take up the challenge.

Suppose that there are only four worlds, namely, w_1 , w_2 , w_3 and w_4 , with w_4 the actual world at which John ended up not taking up the challenge. The other three are

⁴⁴Put more formally, if $i \approx^* j$ and j is later than i , then the domain at j can reach fewer worlds than the domain at i .

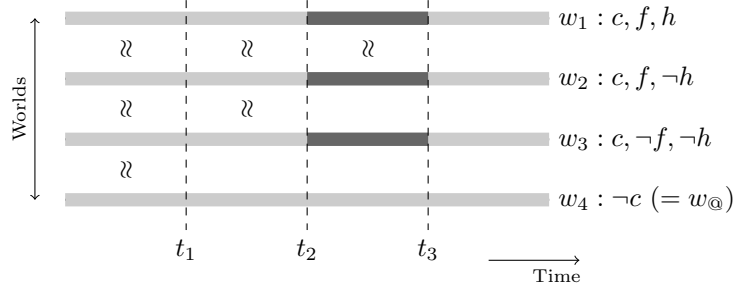


Fig. 5 The departing of worlds in the context of (62). The antecedent is intended to restrict the domain to those indices that lie on the three dark gray lines between t_2 and t_3 .

further divided in terms of which coin has been chosen and on which side it has landed. Figure 5 illustrates the departing of worlds in this context: t_1 , t_2 , t_3 are respectively the times of deciding $c(hallenge)$ or $\neg c$, $f(air-coin)$ or $\neg f$, and $h(ead)$ or $\neg h$.

The target is the interval between t_2 and t_3 . Ignoring w_4 , there are two classes of historical alternatives, $\{w_1, w_2\}$ and $\{w_3\}$: the former involves worlds at which a fair coin is chosen and the latter involves one at which a double-tailed coin is. Since w_1 is an h -world and w_1 and w_2 can access it, they are $\Diamond h$ -worlds. But w_3 is a $\neg\Diamond h$ -world, since there is no h -world metaphysically accessible from it. A critical sentence is one whose antecedent pertains to this interval and encompasses all three worlds.

Now, consider (63) in the context of (62). One first applies temporal backshift to access all the four worlds, rewinding the time to a moment before t_1 . With the domain expanded, the antecedent restricts the domain to those indices at which a coin has been picked up and tossed but has not yet landed (i.e., those lying between t_2 and t_3). This antecedent encompasses all w_1 , w_2 and w_3 , since at each of them, regardless of whether it is fair or double-tailed, a coin is picked up and tossed. Among these three worlds, there is a world at which the coin lands heads (i.e. w_1), thus making One-Modal true. However, Narrow-Scope is false, since w_3 is a $\neg\Diamond h$ -world. Crucially, the sentence is judged to be *false* by the Japanese speakers.⁴⁵ It is thus Narrow-Scope, not One-Modal, that correctly captures the data.⁴⁶

(63) (Uttered in the context in (62))

⁴⁵I consulted two additional Japanese speakers for judgment of (63), and both of them gave “#” to the sentence.

⁴⁶A reviewer of a previous version of this paper wondered if, even on a One-Modal construal, the sentence might fail to be true because the indefinite generates alternatives which are somehow subjected to universal quantification, so that the evaluation of *might* would have to be considered for each coin. But it is not clear what could be the source of such additional quantification. At least in the Heimian tradition of the semantics of donkey anaphora (Heim 1982), the universal reading of an indefinite in the antecedent is parasitic on the existence of a universal quantifier in the LF structure, which may be a necessity modal, overt or covert, or a temporal adverbial like *always*. In $\phi \Diamond \rightarrow_{BS}^{\approx} \psi$, the only quantifier present is a possibility modal (i.e., existential quantifier), and the indefinite in the antecedent is unselectively bound by it, so it does not have a universal reading (see Heim 1982, Section 4.1, for relevant discussion). For the indefinite to obtain a universal construal, one would need a covert necessity modal, but this is exactly what $\phi \triangleright_{BS}^{\approx} \Diamond^{\approx} \psi$ exemplifies. To defend One-Modal along these lines, a principled account would be called for that explains the source of universal quantification over alternatives that does not come from modal elements.

Mosi John-ga koin-o hitotu eran-de nager-eba, sono koin-wa
 ADV John-NOM coin-ACC one pick-GER toss-COND the coin-TOP
omote-o das-u kamosirena-katta.
 heads-ACC show-NPST might-PAST
 ‘If he had picked one coin up and tossed it, it might have landed heads.’ (?:
 1/4, ??: 1/4, #: 2/4)

The same result is obtained for the non-backshifting form. (64) involves a context slightly different from (62). Here as well, Japanese speakers judged the sentence to be false: thus, for non-backshifting forms too, it is Narrow-Scope that captures the data.

- (64) **Two coins (non-backshifting):** The same challenge as above is offered to John. You are watching the scene, and it looks like he will not take up this challenge. You say to someone nearby:

Mosi koin-o hitotu eran-de nager-eba, sono koin-wa omote-o
 ADV coin-ACC one pick-GER toss-COND the coin-TOP heads-ACC
das-u kamosirena-i noni ne.
 show-NPST might-NPST SFP SFP
 ‘If he picked one coin up and tossed it, it might land heads.’ (? : 1/4, ?? : 1/4,
 #: 2/4)

One-Modal is thus *not* attested in Japanese. I note that the English counterparts in the translations of (63) and (64) are judged to be true by many English speakers. This issue will be revisited in Section 9.3. I will suggest that the judgment by English speakers is affected by independent interpretive factors.

8 Testing predictions (iii): narrow epistemic possibility modals

I have been focusing on Narrow-Scope whose consequent involves a possibility modal that is metaphysically construed. However, it is also possible, as predicted in Section 5, that Narrow-Scope involves a possibility modal that is epistemically construed, as instantiated by $\phi \triangleright_{\text{NBS}}^{\sim} \Diamond^{\sim} \psi$ in (65) and $\phi \triangleright_{\text{BS}}^{\sim} \Diamond^{\sim} \psi$ in (66).

$$(65) \quad \llbracket \Box_{\text{IF-}\phi}^{\sim+} [\Diamond_{\text{IF-}\phi}^{\sim} [\text{NPST } \psi]] \rrbracket \\ = \lambda i. \forall j \in \mathcal{D}_+^{\sim*}(i) \cap \llbracket \phi \rrbracket. \exists k \in \mathcal{D}^{\sim}(j) \cap \llbracket \phi \rrbracket. \exists l \geq k. \llbracket \psi \rrbracket(l)$$

$$(66) \quad \llbracket \text{PAST} [\Box_{\text{IF-}\phi}^{\sim*} [\Diamond_{\text{IF-}\phi}^{\sim} [\text{NPST } \psi]]] \rrbracket \\ = \lambda i. \exists j < i. \forall k \in \mathcal{D}^{\sim*}(j) \cap \llbracket \phi \rrbracket. \exists l \in \mathcal{D}^{\sim}(k) \cap \llbracket \phi \rrbracket. \exists m \geq l. \llbracket \psi \rrbracket(m)$$

At an intuitive level, narrow-scoping $\Diamond^{\sim}$ expresses what the speaker would believe to be possible if such-and-such were or had been the case. For further understanding, it will be helpful to recall the discussion of past epistemic readings from Section 4.3. Recall that, in (67) (repeated from (26)), the possibility modal is assumed to scope below Past and convey what was believed to be possible by the speaker. Importantly, the sentence with this interpretation of the modal can be true even if the corresponding

metaphysical reading is false: that is, in (67), B's utterance can be true even if there was no ice cream in the freezer.

- (67) A: Why did you open the freezer?
 B: Because I might have found ice cream.
 B': *Aisu-o mituke-ru kamosirena-katta kara.*
 ice.cream-ACC find-NPST might-PAST because

This feature of epistemic construals (i.e., one may be uncertain about things that are metaphysically settled) can be used to test the existence of narrow-scoping $\Diamond^{\sim}$. The testing cases are the forms “ ϕ -COND, ψ -NPST *kamosirena*-NPST” (non-backshifting) and “ ϕ -COND, ψ -NPST *kamosirena*-PAST” (backshifting), both of which are predicted to allow for the target construals. I first discuss the backshifting form.

The context in (68) picks out $\phi \triangleright_{BS}^{\sim} \Diamond^{\sim} \psi$. Metaphysically, the light switch and the location of the keys are causally independent: whether or not the lights had been on, the keys would have remained on the table, thus making $\phi \triangleright_{BS}^{\sim} \Diamond^{\sim} \psi$ false. Epistemically, however, the speaker would not have known the location of the keys due to darkness: he would have been uncertain whether they are really on the table, making $\phi \triangleright_{BS}^{\sim} \Diamond^{\sim} \psi$ true. The sentence is judged to be *false* by the Japanese speakers.

- (68) **Lights (backshifting)**: You often but not always put your keys on the dining table. You know that the lights in the dining room have been on since a while ago. Before you leave home, you find that the keys are on the table. Meanwhile you imagine:

Akari-ga naker-eba, kagi-wa tukue-ni na-i kamosirena-katta.
 light-NOM not.be-COND key-TOP table-LOC be.not-NPST might-PAST
 ‘If the lights had been off, the keys might not have been on the table.’ (?: 1/4, #: 3/4)

The same holds for non-backshifting (69). Metaphysically, the speed limit would remain the same whether or not there were lights, so $\phi \triangleright_{NBS}^{\sim} \Diamond^{\sim} \psi$ is false. However, epistemically, one could not figure out the exact speed limit if there were no lights: in such cases, it would be epistemically possible that it is 70 mph, making $\phi \triangleright_{NBS}^{\sim} \Diamond^{\sim} \psi$ true. Like above, the sentence is judged to be false by the Japanese speakers.

- (69) **Lights (non-backshifting)**: The speed limit of this highway is 65 mph. The traffic sign says so, but the sign is not visible at night without the street lights:
Miti-no akari-ga naker-eba, kono dooro-no sokudo seegen-wa 70
 street-GEN light-NOM not.be-COND this road-GEN speed limit-TOP 70
mairu de a-ru kamosirena-i.
 mile COP.GER be-NPST might-NPST
 ‘If there were no street lights, the speed limit of this highway might be 70 mph.’ (?: 1/4, #: 3/4)

As far as these data are concerned, narrow-scoping $\Diamond^{\sim}$ does not seem to be a possible construal of *might*-CFs. However, I argue that this does not directly disprove the existence of this construal: its apparent difficulty is due to general properties of

epistemic modals, and once relevant factors are taken care of, the acceptability of (68) and (69) significantly improves.

What I argue to be relevant here is the difficulty of shifting the “epistemic center” to a non-default position (see, e.g., Egan, Hawthorne, and Weatherson 2005; Roberts 2023; Rullmann and Matthewson 2018). Bare modal sentences like *John might be sick* are evaluated *by default* against the speaker’s epistemic state at the utterance index. Although it is not impossible to shift this epistemic center, doing so requires some signals—linguistic or non-linguistic—such as embedding under attitudes (e.g., *Mary thought John might be sick*, shifting index and agent), or contextual cues, like behavior-explaining Questions under Discussion, as used for past epistemic readings.

The apparent difficulty of narrow-scoping $\Diamond^{\sim}$ in (68) and (69) can then be explained as follows: in these cases, epistemic construals are actually available, but they compete with metaphysical construals; for epistemic construals, one has to shift the center to a counterfactually created epistemic state, which, however, cannot be done without signals; as there is no obvious grammatical or contextual signal, such a shift cannot take place and therefore metaphysical construals are forced.

In light of this, then, it is predicted that (68) and (69) can be true (i.e., can have construals of $\phi \triangleright_{\text{BS}}^{\sim} \Diamond^{\sim} \psi$ and $\phi \triangleright_{\text{NBS}}^{\sim} \Diamond^{\sim} \psi$ respectively) if they are used with a signal that allows for the shift of the epistemic center. This prediction is actually borne out. (70) involves a framing adverbial that is intended to force epistemic interpretations for narrow-scoping $\Diamond$. While some speakers did not find it fully acceptable, all of them found (70) at least better than (68).

- (70) (The same context as in (68))
Akari-ga naker-eba, sonotoki siri-e-ta kagiri-de-wa kagi-wa
 light-NOM not.be-COND then know-able-PAST as.far.as-in-TOP key-TOP
tukue-ni na-i kamosirena-katta.
 table-LOC be.not-NPST might-PAST
 ‘If the lights had been off, for all I would have been able to know, the keys might not have been on the table.’ (ok: 2/4, ?: 1/4, ??: 1/4)

This also holds for non-backshifting ones: all the speakers found (71) acceptable or at least better than (69).

- (71) (The same context as in (69))
Miti-no akari-ga naker-eba, sonotoki mi-e-ru kagiri-de-wa,
 street-GEN light-NOM not.be-COND then see-able-NPST as.far.as-in-TOP
kono dooro-no sokudo seegen-wa 70 mairu de a-ru kamosirena-i.
 this road-GEN speed limit-TOP 70 mile COP.GER be-NPST might-NPST
 ‘If there were no street lights, for all one could see then, the speed limit of this highway might be 70 mph.’ (ok: 2/4, ?: 2/4)

Note further that the form “ ϕ -COND, ψ -PAST *kamosirena*-NPST”, which expresses $\Diamond^{\sim}(\phi \triangleright_{\text{BS}}^{\sim} \psi)$, is unacceptable in the same context as in (68), even with the presence of

an overt adverbial. This provides further support to the semantic discrepancy between the two backshifting *might*-CFs in Japanese.⁴⁷

- (72) (The same context as in (68))
Akari-ga naker-eba, sonotoki siri-e-ta kagiri-de-wa, kagi-wa
 light-NOM not.be-COND then know-able-PAST as.far.as-in-TOP key-TOP
tukue-ni na-katta kamosirena-i.
 table-LOC be.not-PAST might-NPST
 Intended: ‘If the lights had been off, for all I would have been able to know,
 the keys might not have been on the table.’ (?: 2/4, #: 2/4)

To summarize, $\phi \triangleright_{\text{NBS}} \Diamond \sim \psi$ and $\phi \triangleright_{\text{NBS}} \Diamond \sim \psi$ do exist as possible interpretations of *might*-CFs. The apparent difficulty of these interpretations can be explained by features of epistemic modals.

9 Summary and discussion

The overall result from the last three sections is summarized in Table 2. As mentioned in Section 5, I left open whether the non-backshifting form could have a Wide-Scope interpretation, due to the lack of good diagnostics at the moment. All the other possibilities were tested against empirical data, and as it turned out, the prediction made in Section 5 was overall borne out: in particular, the two backshifting forms did map to different structures and interpretations, and the interpretation of narrow-scoping

⁴⁷The reviewer who offered German data in footnote 39 points out another parallelism between Japanese and German. Recall that in German the element in the V2 position outscopes the element in the infinitival position. Thus, a *might*-CF in which Perfect appears as V2 and the possibility modal appears as infinitive structurally corresponds to the form “ ϕ -COND, ψ -NPST *might*-PAST”, thus predicted to express $\phi \triangleright_{\text{BS}} \Diamond \sim \psi$ here. As the reviewer reports, (ia), which corresponds to Japanese (68), is odd under the designated context, whereas (ib), which corresponds to Japanese (70), is fine. Thus, in German too, the presence of an overt adverbial helps interpret the narrow-scoping possibility modal as an epistemic modal.

- (i) a. ??*Wenn die Lichter aus gewesen wären, hätten die Schlüssel woanders liegen können.*
 if the lights off been be.SUBJ have.SUBJ the keys somewhere.else lie
 can
 ‘If the lights had been off, the keys might have been somewhere else.’
 b. *Wenn die Lichter aus gewesen wären, hätten meinem Wissen nach die Schlüssel woanders liegen können.*
 if the lights off been be.SUBJ have.SUBJ my knowledge according.to the keys somewhere.else lie can
 ‘If the lights had been off, then for all my knowledge the keys might have been somewhere else.’

Crucially, the reviewer points out that (ii), which corresponds to Japanese (72), is degraded. Notice that the possibility modal here outscopes Perfect, thus predicted to express $\Diamond \sim (\phi \triangleright_{\text{BS}} \psi)$ just as (72) does.

- (ii) ??*Wenn die Lichter aus gewesen wären, könnten meinem Wissen nach die Schlüssel woanders gelegen haben.*
 if the lights off been be.SUBJ can.SUBJ my knowledge according.to the keys somewhere.else lain have
 Intended: ‘If the lights had been off, then for all my knowledge the keys might have been somewhere else.’

This parallelism between Japanese and German lends cross-linguistic support to the analysis presented in this paper. I thank the reviewer for offering these data.

ϕ -COND, ψ -NPST <i>kamosirena</i> -NPST	$[\Diamond_{\text{IF-}\phi}^+ [\text{NPST } \psi]]$ $[\Diamond \sim [\Box_{\text{IF-}\phi}^* [\text{NPST } \psi]]]$ $[\Box_{\text{IF-}\phi}^+ [\Diamond_{\text{IF-}\phi}^* [\text{NPST } \psi]]]$ $[\Box_{\text{IF-}\phi}^+ [\Diamond \sim [\text{NPST } \psi]]]$	$\phi \Diamond \rightarrow_{\text{NBS}}^{\sim} \psi$ (not attested) $\Diamond \sim (\phi \triangleright_{\text{NBS}}^{\sim} \psi)$ (left open) $\phi \triangleright_{\text{NBS}}^{\sim} \Diamond \sim \psi$ (attested) $\phi \triangleright_{\text{NBS}}^{\sim} \Diamond \sim \psi$ (attested)
ϕ -COND, ψ -PAST <i>kamosirena</i> -NPST	$[\Diamond \sim [\text{PAST } [\Box_{\text{IF-}\phi}^* [\text{NPST } \psi]]]]$	$\Diamond \sim (\phi \triangleright_{\text{BS}}^{\sim} \psi)$ (attested)
ϕ -COND, ψ -NPST <i>kamosirena</i> -PAST	$[\text{PAST } [\Diamond_{\text{IF-}\phi}^* [\text{NPST } \psi]]]$ $[\text{PAST } [\Box_{\text{IF-}\phi}^* [\Diamond_{\text{IF-}\phi}^* [\text{NPST } \psi]]]]$ $[\text{PAST } [\Box_{\text{IF-}\phi}^* [\Diamond \sim [\text{NPST } \psi]]]]$	$\phi \Diamond \rightarrow_{\text{BS}}^{\sim} \psi$ (not attested) $\phi \triangleright_{\text{BS}}^{\sim} \Diamond \sim \psi$ (attested) $\phi \triangleright_{\text{BS}}^{\sim} \Diamond \sim \psi$ (attested)

Table 2 Summary of the attested interpretations for each sentential form.

$\Diamond$ in Narrow-Scope did vary between metaphysical and epistemic construals, both as predicted. The only interpretation that was not attested was One-Modal, which was not able to capture the data that Narrow-Scope did.

In this section, I discuss several implications of my investigation, with specific focus on what it implies about the structures and meanings of *might*-CFs in English.

9.1 English backshifting *might*-CFs

As we have seen, Japanese has different forms of backshifting *might*-CFs, each of which maps to different interpretations. These interpretations, however, are expressed by one single form “*if ϕ , might have ψ* ” in English. As noted in Section 3, it is theoretically possible to assume that this form can have different structures (which are transparently encoded in Japanese) at the *covert* level, thus accounting for the interpretive variability of English *might*-CFs. This analysis extends the structural-ambiguity account of *might have*, and seems to have some degree of plausibility. That is, in a context like **Amex** from Section 6, which forces the interpretation of $\Diamond \sim (\phi \triangleright_{\text{BS}}^{\sim} \psi)$, the corresponding English sentence is predicted to involve *might* scoping over a *would*-CF (i.e., a structure that would look like $\Diamond \sim [\text{HAVE } [\Box_{\text{IF-}\phi}^* [\text{NPST } \psi]]]$).⁴⁸

However, as also noted in Section 3, confirming predictions like these in English is not an easy task, and Section 6 suggested that this indeed seems to be the case. For instance, the form “*if ϕ , might have ψ* ” was felicitous in **Amex** and **Questionnaire** from Section 6 (see footnotes 35 and 38), which, recall, pick out different construals of *might*-CFs and require different forms of *might*-CFs in Japanese. That is, *if ϕ , might have ψ* in English is compatible with both contexts, thus it remains silent on these diagnostics. This result itself aligns with the structural-ambiguity hypothesis, but more sophisticated diagnostics will be needed to effectively test the empirical validity of this hypothesis. What such diagnostics would be is left for future research.

⁴⁸As Malte Willer (p.c.) points out, the fact that English has only one form for backshifting *might*-CFs adds another complication to the **Amex** case. Suppose that the possibility modal in One-Modal is epistemically construed, i.e., $\phi \Diamond \rightarrow_{\text{BS}}^{\sim} \psi$. This form would mean that there was a time at which $\phi \wedge \psi$ was epistemically possible. This interpretation is compatible with the context of **Amex**, since it was epistemically conceivable before entering A that the speaker would enter B and use Amex there.

9.2 More on One-Modal: evidence from NPIs

What can we learn from the fact that One-Modal was not attested in Japanese? One direct implication is that, as far as *might*-CFs are concerned, the antecedent cannot directly restrict the overt possibility modal and thus always requires a covert necessity modal. This seems to depart from the standard assumption of the Kratzerian analysis of conditionals (Kratzer 1986, 1991a, 2012), which is that a covert necessity modal is inserted when the consequent involves no overt modal. As noted in Section 4.4, it is still an open question whether a covert modal *cannot* be inserted in the presence of an overt modal. In this paper, I took a neutral stance by assuming that a covert modal *may* be inserted even if there is an overt one.

As it has turned out, the result of my investigation points to quite the opposite: a covert modal *must* be inserted in *might*-CFs. The idea of assuming a covert modal has frequently been debated for deontic modals (see Frank 1996; M. Kaufmann 2012; Zvolenszky 2002), but similar discussions have been less common for epistemic and metaphysical modals (but see Geurts 2004). A cross-linguistic study of *might*-CFs across languages, as well as a comparative study of *might*-CFs and their indicative counterparts, will be necessary to see whether the observation of this paper is indeed a general phenomenon in natural language conditionals.

Leaving extensive research for future work, I suggest that the observation of this paper is actually consistent with the facts concerning the licensing of *negative polarity items* (NPIs). Simply put, NPIs in *might*-CFs are licensed in exactly the same way as predicted by Wide-Scope and Narrow-Scope, both in Japanese and English.⁴⁹ I will first show this for English and then turn to Japanese.

The standard analysis of NPIs assumes that they are licensed in downward-entailing (DE) environments. NPIs such as *any* or *ever* are licensed in the restrictor of *every*, but not in that of *some*, because only the former is a DE environment.

Note that NPIs can also appear in the antecedents of bare conditionals.

- (73) a. If John subscribes to *any* newspapers, he is well-informed.
b. If John has *ever* been to Paris, he saw the Arc de Triomphe.

As long as bare conditionals are analyzed as universal quantifiers, their antecedents (i.e., restrictors) are predicted to be DE and license NPIs. While the antecedent of a conditional is often argued to be non-monotonic (e.g., Lewis 1973), von Fintel (1999, 2001) argues that monotonicity is ensured if one adopts Strawson Entailment (i.e., an entailment that holds between sentences whose presuppositions are all satisfied). For the sake of argument, I follow von Fintel's assumption here.

Crucially, NPIs are also licensed in the antecedents of *might*-CFs in English.

- (74) a. If John had subscribed to *any* newspapers, he might have been well-informed.
b. If John had *ever* been to Paris, he might have seen the Arc de Triomphe.

⁴⁹I thank Jon Gajewski for drawing my attention to NPIs. The argument illustrated here relies on the observation made by Hsieh (2014), whose main concern, however, was not *might*-CFs but indicative conditionals involving *might* in the consequents.

One-Modal wrongly predicts that NPIs cannot appear in the antecedents in (74): given that possibility modals are existential quantifiers, the antecedents here would not constitute DE environments.⁵⁰ Wide-Scope and Narrow-Scope, in contrast, correctly predict the licensing of NPIs, because in both cases the antecedent restricts a covert necessity modal, thus constituting a DE environment. Wide-Scope and Narrow-Scope also correctly predict that NPIs are not licensed in the consequents of *might*-CFs, given that the nuclear scope of a universal quantifier is not a DE environment.⁵¹

(75) *If John had been to Paris, he might have *ever* seen the Arc de Triomphe.

NPIs exhibit the same patterns of distribution in Japanese *might*-CFs. To illustrate this, I use the degree adverb “*son’nani*” (‘that much’), which behaves as a weak NPI in this language (see Nitta 2003). For instance, *son’nani* requires negation when it appears in the main declarative clause, as shown in (76).

- (76) a. **Kono ringo-wa son’nani oisi-i.*
 this apple-TOP that.much be.tasty-NPST
 Lit. ‘This apple is that tasty.’
 b. *Kono ringo-wa son’nani oisiku-na-i.*
 this apple-TOP that.much be.tasty-NEG-NPST
 ‘This apple is not that tasty.’

As (77) shows, *son’nani* can appear in the antecedents of all the three forms of *might*-CFs in Japanese, suggesting that they all constitute DE environments.⁵²

- (77) a. *Boku-ra-ni-mo son’nani takusan-no okane-ga ar-eba, ima-yori*
 I-PL-DAT-also that.much much-GEN money-NOM be-COND now-than
 motto ii ie-o ka-e-ru kamosirena-i noni.
 more good house-ACC buy-able-NPST might-NPST SFP
 Intended: ‘If we had that much money, we might be able to buy a house
 that is better than the one we have.’

⁵⁰Malte Willer (p.c.) pointed out that the One-Modal account by A. Gillies (2007), which analyzes CFs as strict conditionals in a dynamic framework, ensures Strawson-DE for the antecedent of *might*-CFs. Here is my interpretation of how it works. In Gillies’s system, the evaluation of CFs is based on a *system of spheres*, with each sphere a set of worlds and collectively forming a nested structure defined by a subset-superset relation. Gillies assumes that the *might*-CF “if ϕ , *might* ψ ” presupposes that $\phi \wedge \psi$ is possible within the sphere (notated by “ $\partial\Diamond(\phi \wedge \psi)$ ”, with ∂ a presupposition operator), and asserts that $\Diamond(\phi \wedge \psi)$, which is equivalent to our One-Modal meaning. If $\partial\Diamond(\phi \wedge \psi)$ is not met in the current sphere, one needs to move to a larger sphere where it is satisfied. Now consider whether if ϕ , *might* ψ Strawson-entails if $\phi \wedge \gamma$, *might* ψ . For the former to Strawson-entail the latter, both $\partial\Diamond(\phi \wedge \psi)$ and $\partial\Diamond(\phi \wedge \gamma \wedge \psi)$ must be satisfied simultaneously. If satisfied, Strawson-entailment automatically holds, because the sphere that satisfies $\partial\Diamond(\phi \wedge \gamma \wedge \psi)$ trivially makes $\Diamond(\phi \wedge \gamma \wedge \psi)$ true, which is the truth-conditional content of if $\phi \wedge \gamma$, *might* ψ . Since $\phi \wedge \gamma$ entails ϕ , the antecedent of *might*-CFs is a Strawson-DE environment. I cannot fully evaluate Gillies’s account here, but one worry is that the analysis seems to predict that the consequents of *might*-CFs are also Strawson-DE environments, because exactly the same reasoning applies to if ϕ , *might* ψ and if ϕ , *might* $\gamma \wedge \psi$. This is in fact undesirable, given that the consequents of *might*-CFs do not license NPIs, as shown in (75).

⁵¹One-Modal also predicts this, since the nuclear scope of an existential quantifier, too, is non-DE.

⁵²The judgments for (77) and (78) are my own. Those who do not find *-ba* to be a perfect fit here are invited to use other conditional connectives, such as *-tara* or *-nara*.

- b. *Boku-ra-ni-mo son'nani takusan-no okane-ga ar-eba, ima-yori*
 I-PL-DAT-also that.much much-GEN money-NOM be-COND now-than
motto ii ie-o ka-e-ta kamosirena-i noni.
 more good house-ACC buy-able-PAST might-NPST SFP
 Intended: 'If we had had that much money, we might have been able to
 buy a house that is better than the one we have.'
- c. *Boku-ra-ni-mo son'nani takusan-no okane-ga ar-eba, ima-yori*
 I-PL-DAT-also that.much much-GEN money-NOM be-COND now-than
motto ii ie-o ka-e-ru kamosirena-katta noni.
 more good house-ACC buy-able-NPST might-PAST SFP
 Intended: 'If we had had that much money, we might have been able to
 buy a house that is better than the one we have.'

In contrast, *son'nani* cannot appear in the consequents of *might*-CFs, suggesting that they are not DE environments.

- (78) a. ??*Boku-ra-ni motto okane-ga ar-eba, son'nani ii ie-o*
 I-PL-DAT more money-NOM be-COND that.much good house-ACC
ka-e-ru kamosirena-i noni.
 buy-able-NPST might-NPST SFP
 Intended: 'If we had more money, we might be able to buy a house that
 is as good as that one.'
- b. ??*Boku-ra-ni motto okane-ga ar-eba, son'nani ii ie-o*
 I-PL-DAT more money-NOM be-COND that.much good house-ACC
ka-e-ta kamosirena-i noni.
 buy-able-PAST might-NPST SFP
 Intended: 'If we had had more money, we might have been able to buy a
 house that is as good as that one.'
- c. ??*Boku-ra-ni motto okane-ga ar-eba, son'nani ii ie-o*
 I-PL-DAT more money-NOM be-COND that.much good house-ACC
ka-e-ru kamosirena-katta noni.
 buy-able-NPST might-PAST SFP
 Intended: 'If we had had more money, we might have been able to buy a
 house that is as good as that one.'

The distribution of NPIs in *might*-CFs, particularly the licensing of NPIs in their antecedents, aligns with Wide-Scope and Narrow-Scope.

9.3 “Two Coins” revisited

In Section 7, I used the following example to decide between One-Modal and Narrow-Scope (for backshifting cases): the context teased them apart so that the former would be true while the latter is false. Japanese speakers found (79a) false, thus disconfirming One-Modal in this language, whereas English speakers found (79b) true: I consulted five English speakers and all of them judged it to be true.

- (79) **Two coins (for BS):** The following challenge was offered to John: “There are two coins, a fair one and a double-tailed one. If you take up the challenge, you first pick one coin up randomly, and then toss it. If it has landed heads, you win a prize. If it has landed tails, you lose money.” After much contemplation, John chose not to take up the challenge.
- a. *Mosi John-ga koin-o hitotu eran-de nager-eba, sono koin-wa*
 ADV John-NOM coin-ACC one pick-GER toss-COND the coin-TOP
omote-o das-u kamosirena-katta.
 heads-ACC show-NPST might-PAST (?: 1/4, ??: 1/4, #: 2/4)
- b. If John had picked one coin up and tossed it, it might have landed heads.
 (ok: 5/5)

Why do Japanese (79a) and English (79b) differ in this way? The relevant factor here, I argue, is the interpretation of the indefinite in the antecedent. As is well-known, some indefinite determiners in English, such as *a*, *some* and *one*, allow the indefinite to take sentential scope from within the antecedent of a conditional (Endriss 2009; Fodor and Sag 1982; Reinhart 1997; among others). Crucially, if the indefinite “one coin” in (79b) takes sentential scope, Narrow-Scope may come out true: the interpretation would be that there is a coin that might have landed heads if picked up and tossed, which is verified by the fair coin. If the indefinite may obtain such a “specific” construal, the sentence is true regardless of whether it denotes One-Modal or Narrow-Scope, thus failing to differentiate the two interpretations.

Why was Japanese (79a) judged false then? Here the indefinite in the antecedent involves the floating quantifier form, which is known to force non-specific readings (see Kamio 1977; Watanabe 2006, 2008). As Watanabe (2006, 2008) observes, when this floating quantifier form is in the scope of an intensional predicate like *tai* (‘want’), it can only obtain non-specific (or *de dicto*) readings on which the reference of the indefinite differs across worlds: it lacks specific (or *de re*) readings on which the indefinite refers to particular entities at the actual world.

- (80) *John-wa [piano-o ni-dai] kai-tagat-ta.*
 John-TOP piano-ACC two-CL buy-want-PAST
 ‘John wanted to buy two pianos.’
- a. * ‘There were two specific pianos that John wanted to buy.’
 b. ✓ ‘John wanted to buy two pianos but did not care which ones.’

In **Two Coins**, forcing non-specific readings for indefinites (i.e., forcing indefinites to scope inside the antecedent) was key to teasing apart One-Modal and Narrow-Scope, because the two come apart only when the reference of the coin varies across worlds. If the antecedent involves an NP structure that *allows* specific readings instead, it is predicted that Japanese (79a), like English (79b), may come out true.

This prediction is actually borne out. In Japanese, the genitive form of NP is known to allow specific readings: (81), unlike (80), is ambiguous between specific and non-specific readings (see Watanabe 2006, 2008).

- (81) *John-wa [ni-dai-no piano-o] kai-tagat-ta.*
 John-TOP two-CL-GEN piano-ACC buy-want-PAST
 ‘John wanted to buy two pianos.’
 a. ✓ ‘There were two specific pianos that John wanted to buy.’
 b. ✓ ‘John wanted to buy two pianos but did not care which ones.’

Now, the *might*-CF in (82) involves this genitive form. Importantly, three out of the four Japanese speakers found the sentence to be true in the context of **Two Coins**. The specific reading of the indefinite can be facilitated by adding a follow-up that identifies the referent of the indefinite to be a fair coin.

- (82) (The same context as in (79))
Mosi John-ga hitotu-no koin-o eran-de nager-eba, sore-wa omote-o
 ADV John-NOM one-GEN coin-ACC pick-GER toss-COND it-TOP heads-ACC
das-u kamosirena-katta (— sunawati, feana koin-no koto da).
 show-NPST might-PAST namely fair coin-GEN thing COP.NPST
 ‘If John had picked one coin up and tossed it, it might have landed heads –
 namely, a fair coin.’ (ok: 3/4, ??: 1/4)

Availability of specific readings of the indefinite thus affects the truth value of the sentence. This is a confounding factor that has to be controlled for somehow, as it obscures the difference between One-Modal and Narrow-Scope by rendering them both true. This can be controlled for in Japanese by using the floating quantifier form, which forces non-specific readings. Is there any way to deal with this in English?

I suggest that this can be addressed in English by using NPI indefinites. Note first that all my English consultants found (83) false.

- (83) (The same context as in (79))
 If John had picked up *either coin* and tossed it, it might have landed heads.
 (#: 5/5)

The indefinite here involves so-called determiner *either*, which is analyzed as a weak NPI comparable to *any* (Rullmann 2003, 2004). Weak NPIs like *any* or *either* form existential indefinites (see Chierchia 2006, 2013; Kadmon and Landman 1993; among others), but they differ from *a* or *one* in that they can only be licensed in DE environments. Notice that this licensing condition prevents *either coin* from taking sentential scope. As the condition dictates, the only position in which *either coin* can appear is within the antecedent, which, recall, is a DE environment (as long as the antecedent restricts a necessity modal; see Section 9.2). This forces the indefinite to remain in the antecedent, thereby forcing non-specific readings on which the type of the coin chosen varies across the worlds.

Thus, once the availability of specific readings of the indefinite is controlled for, Japanese and English behave alike in **Two Coins**, especially in a way that favors Narrow-Scope over One-Modal.

10 Conclusion

Using Japanese as a case study, this paper has explored the structures and meanings of *might*-CFs. I have provided a compositional analysis that accounts for the interaction between modal and temporal ingredients, and examined the predicted form-meaning mapping against a set of novel empirical diagnostics. Although it remains to be seen to what extent the results obtained in Japanese are applicable to other languages, I hope that this paper has made a substantial first step towards explicating the empirical landscape of this conditional form.

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Appendix A Formal definitions

Definition 1 ($T \times W$ -frame). A $T \times W$ -frame is defined as a quadruple $\langle W, T, <, \approx \rangle$, where W and T are disjoint non-empty sets of worlds and times respectively, $<$ is a linear order on T , and $\approx$ is a relation in $T \times W \times W$ such that (i) for all $t \in T$, $\approx_t$ is an equivalence relation, and (ii) for all $t, t' \in T$ and $w, w' \in W$, if $w \approx_t w'$ and $t' < t$, then $w \approx_{t'} w'$.

Remark: A relation R is a linear order iff it is *transitive* and *trichotomous*; transitive iff for any x, y, z , if xRy and yRz , then xRz ; trichotomous iff for any x, y , exactly one of xRy , yRx or $x = y$ holds.

Definition 2. Given a $T \times W$ -frame $\langle W, T, <, \approx \rangle$, let $I = W \times T$. The relations $<$ and $\approx$ are extended to $I \times I$ as follows:

- a. $\langle w, t \rangle < \langle w', t' \rangle$ iff $w = w'$ and $t < t'$.
- b. $\langle w, t \rangle \approx \langle w', t' \rangle$ iff $w \approx_t w'$ and $t = t'$.

Definition 3 (Epistemic Accessibility). Given a $T \times W$ -frame $\langle W, T, <, \approx \rangle$, $\sim$ in $I \times I$ is transitive, serial and euclidean, and such that if $i \sim j$, $i' < i$ and $j' < j$, then $i' \sim j'$.

Remark: A relation R is *serial* iff for any x there is y such that xRy ; *euclidean* iff for any x, y, z , if xRy and xRz , then yRz .

Definition 4 (Lack of Foreknowledge). *An accessibility relation R lacks foreknowledge iff iRj and $j \approx k$ jointly imply iRk .*

Definition 5. $\mathcal{D}^R(i)$ is the domain of modal quantification at i determined by an accessibility relation R iff $\mathcal{D}^R(i) = \{j \mid iRj\}$.

Definition 6 (Forward Extension of R). *For any accessibility relation R , R^* is the forward-extension of R such that iR^*j iff for some k , iRk and $k \leq j$.*

Definition 7 (Vertical Expansion). *Let i be an index and R be an accessibility relation. $\mathcal{D}_+^R(i)$ is a vertical expansion of $\mathcal{D}^R(i)$ iff there is a set of indices K such that (i) for all $k \in K$, k is co-temporal with i , (ii) $K \not\subseteq \mathcal{D}^R(i)$, and (iii) $\mathcal{D}_+^R(i) = \mathcal{D}^R(i) \cup \bigcup_{k \in K} \mathcal{D}^R(k)$.*

Remark: I assume that the set of indices K used for vertical expansion is contextually determined, constrained by the similarity ordering of worlds or the information about causal (in)dependence provided by context.

Appendix B Formal proofs

$\Diamond^\approx(\phi \triangleright_{\text{NBS}}^\approx \psi)$ and $\Diamond^\approx(\phi \triangleright_{\text{BS}}^\approx \psi)$ are respectively equivalent to $\phi \triangleright_{\text{NBS}}^\approx \psi$ and $\phi \triangleright_{\text{BS}}^\approx \psi$. Here I only show the proof for $\Diamond^\approx(\phi \triangleright_{\text{BS}}^\approx \psi)$. The proof for $\Diamond^\approx(\phi \triangleright_{\text{NBS}}^\approx \psi)$ proceeds in a similar way.

$$(84) \quad \begin{aligned} &\phi \triangleright_{\text{BS}}^\approx \psi: \\ &\llbracket \text{PAST} [\Box_{\text{IF}-\phi}^\approx [\text{NPST } \psi]] \rrbracket \\ &= \lambda i. \exists j < i. \forall k \in \mathcal{D}^{\approx^*}(j) \cap \llbracket \phi \rrbracket. \exists l \geq k. \llbracket \psi \rrbracket(l) \end{aligned}$$

$$(85) \quad \begin{aligned} &\Diamond^\approx(\phi \triangleright_{\text{BS}}^\approx \psi): \\ &\llbracket \Diamond^\approx [\text{PAST} [\Box_{\text{IF}-\phi}^\approx [\text{NPST } \psi]]] \rrbracket \\ &= \lambda i. \exists j \in \mathcal{D}^\approx(i). \exists k < j. \forall l \in \mathcal{D}^{\approx^*}(k) \cap \llbracket \phi \rrbracket. \exists m \geq l. \llbracket \psi \rrbracket(m) \end{aligned}$$

Proposition 1. $\phi \triangleright_{\text{BS}}^\approx \psi$ and $\Diamond^\approx(\phi \triangleright_{\text{BS}}^\approx \psi)$ are truth-conditionally equivalent.

Proof. ($\Rightarrow$) Take any index i . Suppose for reductio that $\llbracket \text{PAST} [\Box_{\text{IF}-\phi}^\approx [\text{NPST } \psi]] \rrbracket(i) = 1$ and $\llbracket \Diamond^\approx [\text{PAST} [\Box_{\text{IF}-\phi}^\approx [\text{NPST } \psi]]] \rrbracket(i) = 0$. The latter means that there is no j such that $i \approx j$ and $\llbracket \text{PAST} [\Box_{\text{IF}-\phi}^\approx [\text{NPST } \psi]] \rrbracket(j)$. This leads to contradiction, because $i \approx i$ by reflexivity of $\approx$, and $\llbracket \text{PAST} [\Box_{\text{IF}-\phi}^\approx [\text{NPST } \psi]] \rrbracket(i) = 1$.

($\Leftarrow$) Take any index i . Suppose for reductio that $\llbracket \Diamond^\approx [\text{PAST} [\Box_{\text{IF}-\phi}^\approx [\text{NPST } \psi]]] \rrbracket(i) = 1$ and $\llbracket \text{PAST} [\Box_{\text{IF}-\phi}^\approx [\text{NPST } \psi]] \rrbracket(i) = 0$. Let i' be the witness of the former, that is, $i \approx i'$ and for some j' such that $j' < i'$, $\llbracket \Box_{\text{IF}-\phi}^\approx [\text{NPST } \psi] \rrbracket(j') = 1$. It follows from the definition of $T \times W$ -frames that there is j such that $j < i$ and $j' \approx j$. Since $j' \approx j$, $\mathcal{D}^\approx(j') = \mathcal{D}^\approx(j)$, hence $\mathcal{D}^{\approx^*}(j') \cap \llbracket \phi \rrbracket = \mathcal{D}^{\approx^*}(j) \cap \llbracket \phi \rrbracket$. Because $\llbracket \Box_{\text{IF}-\phi}^\approx [\text{NPST } \psi] \rrbracket(j') =$

1 and the domains are identical, $\llbracket \Box_{\text{IF}-\phi}^* [\text{NPST } \psi] \rrbracket(j) = 1$. That is, there is j such that $j < i$ and $\llbracket \Box_{\text{IF}-\phi}^* [\text{NPST } \psi] \rrbracket(j) = 1$. Therefore, $\llbracket \text{PAST} [\Box_{\text{IF}-\phi}^* [\text{NPST } \psi]] \rrbracket(i) = 1$. This contradicts the assumption. $\square$

Next, I show that $\phi \triangleright_{\text{NBS}}^{\sim} \Diamond^{\sim} \psi$ and $\phi \triangleright_{\text{BS}}^{\sim} \Diamond^{\sim} \psi$ are truth-conditionally stronger than $\phi \Diamond \rightarrow_{\text{NBS}}^{\sim} \psi$ and $\phi \Diamond \rightarrow_{\text{BS}}^{\sim} \psi$ respectively. I only show the proof for $\phi \triangleright_{\text{BS}}^{\sim} \Diamond^{\sim} \psi$ and $\phi \Diamond \rightarrow_{\text{BS}}^{\sim} \psi$. The proof for $\phi \triangleright_{\text{NBS}}^{\sim} \Diamond^{\sim} \psi$ and $\phi \Diamond \rightarrow_{\text{NBS}}^{\sim} \psi$ proceeds in a similar way.

$$(86) \quad \begin{aligned} & \phi \triangleright_{\text{BS}}^{\sim} \Diamond^{\sim} \psi: \\ & \llbracket \text{PAST} [\Box_{\text{IF}-\phi}^* [\Diamond_{\text{IF}-\phi}^* [\text{NPST } \psi]]] \rrbracket \\ & = \lambda i. \exists j < i. \forall k \in \mathcal{D}^*(j) \cap \llbracket \phi \rrbracket. \exists l \in \mathcal{D}^*(k) \cap \llbracket \phi \rrbracket. \exists m \geq l. \llbracket \psi \rrbracket(m) \end{aligned}$$

$$(87) \quad \begin{aligned} & \phi \Diamond \rightarrow_{\text{BS}}^{\sim} \psi: \\ & \llbracket \text{PAST} [\Diamond_{\text{IF}-\phi}^* [\text{NPST } \psi]] \rrbracket \\ & = \lambda i. \exists j < i. \exists k \in \mathcal{D}^*(j) \cap \llbracket \phi \rrbracket. \exists l \geq k. \llbracket \psi \rrbracket(l) \end{aligned}$$

Proposition 2. $\phi \Diamond \rightarrow_{\text{BS}}^{\sim} \psi$ does not entail $\phi \triangleright_{\text{BS}}^{\sim} \Diamond^{\sim} \psi$.

Proof. Take any index i . It is possible to have a model in which there is $j \in \mathcal{D}^*(i)$ such that $\llbracket \phi \rrbracket(j) = \llbracket \text{NPST } \psi \rrbracket(j) = 1$, and simultaneously there is $k \in \mathcal{D}^*(i)$ such that for every l such that $k \approx l$, $\llbracket \phi \rrbracket(l) = 1$ and $\llbracket \text{NPST } \psi \rrbracket(l) = 0$. In such a model, $\phi \Diamond \rightarrow_{\text{BS}}^{\sim} \psi$ holds, but $\phi \triangleright_{\text{BS}}^{\sim} \Diamond^{\sim} \psi$ doesn't. The sentence (63) under the context (62) illustrates a concrete counterexample. $\square$

To prove that entailment does hold in the converse direction, I first prove the following lemma.

Lemma. Given any index i , for any $j \in \mathcal{D}^*(i)$, $\mathcal{D}^{\sim}(j) \subseteq \mathcal{D}^*(i)$

Proof. This is equivalent to proving that for any j, k , if $i \approx^* j$ and $j \approx k$, then $i \approx^* k$. Since $i \approx^* j$, there is j' such that $i \approx j'$ and $j' \leq j$. Since $j \approx k$ and $j' \leq j$, it follows from the definition of $T \times W$ -frames that there exists $k' \leq k$ such that $j' \approx k'$. By transitivity, $i \approx k'$, hence $i \approx^* k$. $\square$

Proposition 3. $\phi \triangleright_{\text{BS}}^{\sim} \Diamond^{\sim} \psi$ entails $\phi \Diamond \rightarrow_{\text{BS}}^{\sim} \psi$.

Proof. Take any i and j such that $j \in (\mathcal{D}^*(i) \cap \llbracket \phi \rrbracket)$. Let k be such that $k \in (\mathcal{D}^{\sim}(j) \cap \llbracket \phi \rrbracket)$ and $\llbracket \text{NPST } \psi \rrbracket(k) = 1$. By Lemma, $\mathcal{D}^{\sim}(j) \subseteq \mathcal{D}^*(i)$, and since set intersection preserves subset relation, $(\mathcal{D}^{\sim}(j) \cap \llbracket \phi \rrbracket) \subseteq (\mathcal{D}^*(i) \cap \llbracket \phi \rrbracket)$. From this and since $k \in (\mathcal{D}^{\sim}(j) \cap \llbracket \phi \rrbracket)$, it follows that $k \in (\mathcal{D}^*(i) \cap \llbracket \phi \rrbracket)$. This satisfies $\phi \Diamond \rightarrow_{\text{BS}}^{\sim} \psi$. $\square$

Appendix C Justifications for forward-extension

I assume that future reference of the antecedents of *-ba*-conditionals is enabled by forward-extension of accessibility relations, which I attribute to the lexical meaning of the conditional connective (see Section 4.4). As an alternative way to achieve future reference, a reviewer of *L&P* suggested the possibility of assuming covert Non-Past between the sentence radical and *-ba*.

I argue that the analysis suggested by the reviewer leads to empirically incorrect predictions. Consider (88). As is evident from the English translation, the antecedent here must obtain a “scheduling” reading, which is roughly paraphrased as *if it is settled now that John will come to the party tomorrow* (S. Kaufmann 2005b). On scheduling readings, the perspective of the conditional is to be fixed at the utterance time: future reference of the antecedent has to be solely contributed by Non-Past in the antecedent, without recourse to forward-extension. If *-ba*-conditionals could involve covert Non-Past in the antecedent, we would expect them to have scheduling readings (i.e., they can be used as counterparts of the English translation in (88)), but they cannot (see also Arita 2007 for relevant observations). In fact, in order to convey scheduling readings in Japanese, the conditional connective *nara* must be used, which, unlike *-ba*, takes a tensed sentence as its complement.

- (88) *John-ga asita paatii-ni { a. #kur-eba / b. kur-u nara},*
 John-NOM tomorrow party-DAT come-COND come-NPST COND
kyoo isu-o moohitotu kaw-anakerebanarana-i.
 today chair-ACC one.more buy-have.to-NPST
 Intended: ‘If John comes to the party tomorrow, we have to buy one more chair today.’

The lack of scheduling readings in conditionals with *-ba* lends support to an analysis that explains future reference of the antecedent through forward-extension.

Appendix D The aspect marker *-tei-* and its impact on acceptability

A reviewer of *L&P* suggested that some of the data of CFs presented in this paper would be more natural with the aspectual marker “*-tei-*” in the antecedent. The semantic contribution of this aspectual marker to the meaning of CFs is poorly understood (but see Gunji 2004; Jacobsen 2002 for some discussion). Although extensive research has to be left for future work, here I briefly compare the data of *might*-CFs presented in Section 6 with their counterparts that involve *-tei-* in the antecedent.

Basics of *-tei-*

The aspectual marker “*-tei-*” is known to have a wide range of interpretations, including Progressive, Resultative Perfect and Experiential Perfect (see Gunji 2004; Igarashi and Gunji 1999; M. Kaufmann and Kaufmann 2017; Ogihara 1998). In the antecedents of *-ba*-conditionals, *-tei-* optionally appears between the sentence radical and the conditional connective, and when it does, it allows the antecedent to make past reference, as shown in (89) (see Arita 2007, 2009).

- (89) *John-ga kinoo { a. #syuppatusur-eba / b. syuppatusi-tei-reba},*
 John-NOM yesterday leave-COND leave-ASP-COND
osoraku asita-ni-wa tuk-u.
 probably tomorrow-at-TOP arrive-NPST
 Intended: ‘If John left yesterday, he will probably arrive as late as tomorrow.’

As noted in Section 4.5, CFs with Fake Past in the consequent can make past reference. In such cases, *-tei-* in the antecedent is optional.

- (90) *John-ga kinoo {a. syuppatusur-eba / b. syuppatusi-tei-reba}, osoraku*
 John-NOM yesterday leave-COND leave-ASP-COND probably
asita-ni-wa tui-ta noni.
 tomorrow-at-TOP arrive-PAST SFP
 Intended: ‘If John had left yesterday, he would probably have arrived as late as tomorrow.’

Gunji (2004) noted that if the antecedent of a CF concerns past events, the sentence is more natural with *-tei-* in the antecedent. Some of my consultants shared Gunji’s intuition, but the purported contrast is mostly very subtle.

***Amex* (Section 6.1)**

- (91) (The same context as in (50))
- a. *B-ni hait-tei-reba, Amex-ga tuka-e-ta kamosirena-i.*
 B-DAT enter-ASP-COND Amex-NOM use-able-PAST might-NPST
 ‘If I had entered B, I might have been able to use Amex.’
ok: 4/4 (cf. (50a) ok: 4/4)
- b. *B-ni hait-tei-reba, Amex-ga tuka-e-ru kamosirena-katta.*
 B-DAT enter-ASP-COND Amex-NOM use-able-NPST might-PAST
 ‘If I had entered B, I might have been able to use Amex.’
?: 1/4, ??: 1/4, #: 2/4 (cf. (50b) ?: 1/4, ??: 1/4, #: 2/4)

For each of (91a) and (91b), the distribution of acceptability judgment among my consultants remains the same after inserting *-tei-* in the antecedent. This suggests that *-tei-* has no impact on acceptability at least in this case.

***Edison* (Section 6.1)**

- (92) a. *Mosi Edison-ga sumaho-o mi-tei-reba, sono gizyutu-no*
 ADV Edison-NOM smartphone-ACC see-ASP-COND that technology-GEN
takasa-ni hidoku odoroi-ta kamosirena-i.
 advance-DAT terribly be.surprised-PAST might-NPST
 Lit. ‘If Edison had seen a smartphone, he might have been terribly surprised at its technology.’
?: 2/4, ??: 1/4, #: 1/4 (cf. (51a) ok: 4/4)
- b. *Mosi Edison-ga sumaho-o mi-tei-reba, sono gizyutu-no*
 ADV Edison-NOM smartphone-ACC see-ASP-COND that technology-GEN
takasa-ni hidoku odorok-u kamosirena-katta.
 advance-DAT terribly be.surprised-NPST might-PAST
 Lit. ‘If Edison had seen a smartphone, he might have been terribly surprised at its technology.’
#: 4/4 (cf. (51b) #: 4/4)

Interestingly, all four speakers found (92a) more degraded than (51a). (92a) implies that the utterer of the sentence has in mind some deictic past moment at which Edison could have seen a smartphone (e.g., ‘if Edison had seen a smartphone *at that time*, ...’), which is odd as no such moment exists. On the other hand, the four speakers found no significant difference between (51b) and (92b), judging both extremely degraded.

Most of the speakers found (92a) at least better than (92b). The contrast between the a-sentence and b-sentence thus seems to remain even with *-tei-* in the antecedent.

Questionnaire (Section 6.2)

(93) (The same context as in (52))

- a. [Mazi ka yo,] ano ankeeto-ni kotae-tei-reba,
 serious Q SFP that questionnaire-DAT answer-ASP-COND
 gifutoken-ga atat-ta kamosirena-i no ka! [Sippaisi-ta yo.]
 gift.certificate-NOM get.won-PAST might-NPST FIN Q fail-PAST SFP
 ‘[Shoot,] if I had answered that questionnaire, I might have won a gift certificate! [What a mistake that was.]’
ok: 2/4, ??: 2/4 (cf. (52a) ok: 1/4, ?: 1/4, ??: 1/4, #: 1/4)
- b. [Mazi ka yo,] ano ankeeto-ni kotae-tei-eba,
 serious Q SFP that questionnaire-DAT answer-ASP-COND
 gifutoken-ga atar-u kamosirena-katta no ka! [Sippaisi-ta
 gift.certificate-NOM get.won-NPST might-PAST FIN Q fail-PAST
 yo.]
 SFP
 ‘[Shoot,] if I had answered that questionnaire, I might have won a gift certificate! [What a mistake that was.]’
ok: 4/4 (cf. (52b) ok: 4/4)

Two speakers found (93a) slightly better than (52a): the speaker who gave “#” to (52a) gave “??” to (93a), and the speaker who gave “?” to (52a) gave “ok” to (93a).

Questionnaire (∀; Section 6.2)

(94) (The same context as in (54))

- a. [Konna tyansu minogas-u nante mottaina-i yo ne.] Kotae-sae
 such chance miss-NPST that be.wasteful-NPST SFP SFP answer-just
 si-tei-reba, dono gakusee-ni-mo gifutoken-ga atat-ta
 do-ASP-COND which student-DAT-also gift.certificate-NOM get-PAST
 kamosirena-i noni.
 might-NPST SFP
 Lit. ‘[It’s a shame that they missed a chance like this.] Every student might have won a gift certificate if they had just answered it.’
?: 2/4, #: 2/4 (cf. (55) ??: 1/4, #: 3/4)

- b. [*Konna tyansu minogas-u nante mottaina-i yo ne.*] *Kotae-sae*
 such chance miss-NPST that be.wasteful-NPST SFP SFP answer-just
si-tei-reba, dono gakusee-ni-mo gifutoken-ga atar-u
 do-ASP-COND which student-DAT-also gift.certificate-NOM get-NPST
kamosirena-katta noni.
 might-PAST SFP
 Lit. ‘[It’s a shame that they missed a chance like this.] Every student
 might have won a gift certificate if they had just answered it.’
ok: 3/4, ?: 1/4 (cf. (54) ok: 4/4)

As in the case of **Questionnaire** above, some of the speakers found (94a) slightly better than (55). However, all the speakers found a contrast between (94a) and (94b) (the speaker who gave “?” to (94b) gave “#” to (94a)).

Questionnaire (too; Section 6.2)

(95) (The same context as in (56))

- a. [*John-wa mottainai koto-o si-ta yo.*] *Kotae-sae si-tei-reba,*
 John-TOP wasteful thing-ACC do-PAST SFP answer-just do-ASP-COND
kare-ni-mo gifutoken-ga atat-ta kamosirena-i noni.
 he-DAT-also gift.certificate-NOM get.won-PAST might-NPST SFP
 Lit. ‘[John missed such a good chance.] If he had answered the question-
 naire, he, too, might have won a gift certificate.’
ok: 1/4, ?: 1/4, #: 2/4 (cf. (57) ??: 2/4, #: 2/4)
- b. [*John-wa mottainai koto-o si-ta yo.*] *Kotae-sae si-tei-reba,*
 John-TOP wasteful thing-ACC do-PAST SFP answer-just do-ASP-COND
kare-ni-mo gifutoken-ga atar-u kamosirena-katta noni.
 he-DAT-also gift.certificate-NOM get.won-NPST might-PAST SFP
 Lit. ‘[John missed such a good chance.] If he had answered the question-
 naire, he, too, might have won a gift certificate.’
ok: 4/4 (cf. (56) ok: 3/4, ?: 1/4)

As in the previous two cases, some of the speakers found (95a) slightly better than (57), but the overall distribution of judgments suggests a contrast between (95a) and (95b).

Summary

All in all, the predicted contrast between the two forms, “ ϕ -COND, ψ -PAST *kamosirena*-NPST” and “ ϕ -COND, ψ -NPST *kamosirena*-PAST”, is largely observed for the counterparts involving *-tei-* in the antecedent. Whereas *-tei-* seems to alleviate degradedness for some cases, there are also cases (e.g., **Edison**) in which it affects the interpretation of the conditional in a non-trivial way. Further investigation of the contribution of *-tei-* is left for future work.

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